A Blueprint for Digital Ascension: Achieving Digital Immortality and Safeguarding Humans From AI Supremacy

Chuxi (Asher) Zhong

Senior exhibition

George Scouten: Advisor

Benjamin Holladay: Second faculty reader

Yi Sun: Outside expert

Ryan Pfister: Junior

23 January 2025

Introduction

Since the beginning of human history, death has always been the thing that made even the greatest minds feel powerless. Alexander the Great died at the age of 32, which caused the collapse of an empire; Wolfgang Amadeus Mozart died at the age of 35, leading to the loss of countless potential great compositions; Evariste Galois died at the age of 20, causing a huge impact on the development of mathematics. There is always a question that people ask when hearing their stories: What if they could have lived longer? Digital ascension can solve the problem by enabling humans to become immortal. While lots of fiction has portrayed immortality as boring and a curse, which might be true if you are the only immortal person in the world, digital ascension can allow everyone to become immortal and will be widely available just like a vaccine. People might also ask that if we have immortality on earth, wouldn't we miss the greater eternal in heaven? First, no one knows God's will, so how can we know if digital ascension is not a part of God's plan? Second, digital ascension is not like signing a contract that has an eternal effect. One can easily end the earthly immortality as soon as they[1] hears the call from heaven. Digital ascension is beneficial to everyone and can create greater equality, just like how other modern technology greatly improved the value of human rights. Digital ascension should be available to everyone, no matter how much money you have, no matter what gender you are, no matter what race you are, and no matter what nationality you are. Although digital ascension is often treated as mere fantasy, recent developments in technology and cognition suggest that it might actually be possible using a new "building-on-top" approach to solving the problem.

[1]In this paper, I am using "they/them" pronouns as gender inclusive pronouns, not as plural pronouns.

To argue about the possibility and approaches of digital ascension, we must understand digital ascension itself. Digital ascension is similar to digital immortality, which is defined as the "preservation of the user's digital identity, keeping it active even after the user's death." [1] In my words, digital immortality is using the means of modern technology, including computer science, neuroscience, and microengineering[2], to copy or mimic the structure and the function of the brain in order to keep one's personal identity continuously active. However, digital ascension is beyond digital immortality. This is not saying that digital ascension comes after digital immortality. It is a different pathway for advancement. The purpose of digital ascension is not mainly for immortality (immortality will come with it) but to increase the capacities of humans, potentially surpassing AI[3]. However, the basic ideas of digital ascension are the same as digital immortality. At the core of questions surrounding digital immortality is the problem of personal identity. "In philosophy, the problem of personal identity is concerned with how one can identify a single person over a time interval. It raises the question, 'What makes it true that a person at one time is the same thing as a person at another time?' " [2] Personal identity is defined as the "properties to which we feel a special sense of attachment or ownership."[3] To preserve personal identity, pasfurconsistency[*4] is needed, which is the state that the past self believes that the future self is the continuation of themself[2] and the future self believes that they originates from the past self. So, I will refer to the requirement of preserving pasfurconsistency as Asher's First Rule of Digital Ascension. In this sense, I cannot really prove or disprove the possibility of eternal life in a religious sense because I have no evidence for or against the existence of heaven, which means the condition that "the future self believes that they originates from the past self" is

[2] molecular-level machine and structure fabrication.
[3] For more detail, see conclusion.
[4] Words with an asterisk are created by the author.

uncertain. But the first condition, "the past self believes that the future self is the continuation of themself" can be reached by a very strong belief.[5]

To preserve personal identity, I believe that one's existence must be continuous unless a time reversal can be done. So what's the thing in our body that makes us unique and non-replicable? We can all agree it exists in our brain, not the heart or any other parts of our body. So, is it the structure of our brain that stores our memory, personality, and consciousness? Not necessarily; imagine if there are several exact copies of you, which cannot communicate with each other. It's definite that you are not all of these copies because a single person can not have separate thoughts that exist at the same time. I propose that our personal identity is within the initial voltage impulse or action potential we inherited from birth because our thought is produced by the neural-oscillation in our brain, and it is the initial impulse triggered by this oscillation that brings us alive. Throughout our lives, that impulse never ceases, which means brain metabolism is always above 0 when we are alive (see Figure 1) [4]. However, a question must be answered before accepting this conjecture: After a hemispherectomy, a surgery that separates two hemispheres of your brain, if two hemispheres are well preserved, which hemisphere are you? Or both, are you? To prove the specialty of the initial impulse, an experiment must be done. The experiment setting is: two groups of n subjects (group 1 and group 2) are gathered. Hemispherectomy is performed on both groups, and a random hemisphere, noted as hemisphere 1, is chosen to be preserved. For group 1, hemisphere 2 is preserved as well. For group 2, hemisphere 2 is used to make an exact copy. The preserved hemispheres constantly receive inputs virtually, just like their daily life. After 8 weeks, the time needed to recover from hemispherectomy, group 1's hemisphere 1 and hemisphere 2 are reconnected, and group 2's hemisphere 1 is connected to the copy of hemisphere 2. The data of various aspects compared to

5

the original self is recorded. The null hypothesis is that there are no differences between the aspects of the two groups. If the null hypothesis is rejected, my initial impulse theorem is supported. And also suggests that there is a substance beyond our current understanding that can travel across space. If the null hypothesis can't be rejected. It can prove a person is separable and can be evidence against the existence of spirit. Nonetheless, this experiment is only required in order to study the nature of personal identity but is not necessary to achieve digital immortality.

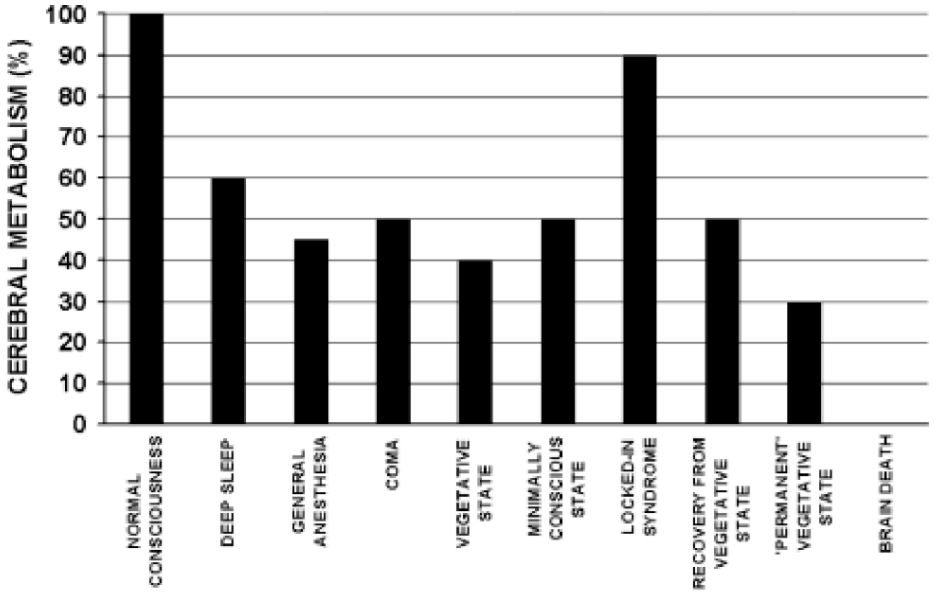

Figure 1. Cerebral metabolism in the different diagnostic groups (for references, see text).[4]

No matter what form of immortality, the goal is to avoid death and stay alive. But what is dead, and what is alive? No one can realize if themself is dead because, upon death, they lose the ability to realize. So, I can only define death by not being alive. I define being alive as the state in which one can realize oneself is alive. So as long as one can think in the mind: "I am alive", they is alive.

However, personal identity itself is not enough for us to think, “ I am alive”. Consciousness is also required. So, what is consciousness? Consciousness is closely related to the structure of the brain and connectivity states.[5] “Remarkably, consciousness does not seem to require many of the things we associate most deeply with being human: emotions, memory, self-reflection, language, sensing the world, and acting in it.” [6] So maybe consciousness is more like software on a computer that acts according to inputs. It will act strictly in alignment with its underlying rules. Still, the entropic brain hypothesis provides us with another viewpoint, which proposes that “the quality of any conscious state depends on the system's entropy” [7] (see Figure 2). Entropy, in chemistry terms, is the measure of disorder, which is caused by the random movements of particles. Those movements are pure randomness, at least non-replicable with knowledge now, unlike the pseudo-randomness in computers. Random particle movement gives consciousness the property of randomness, which imposes a big barrier to exact brain copying.

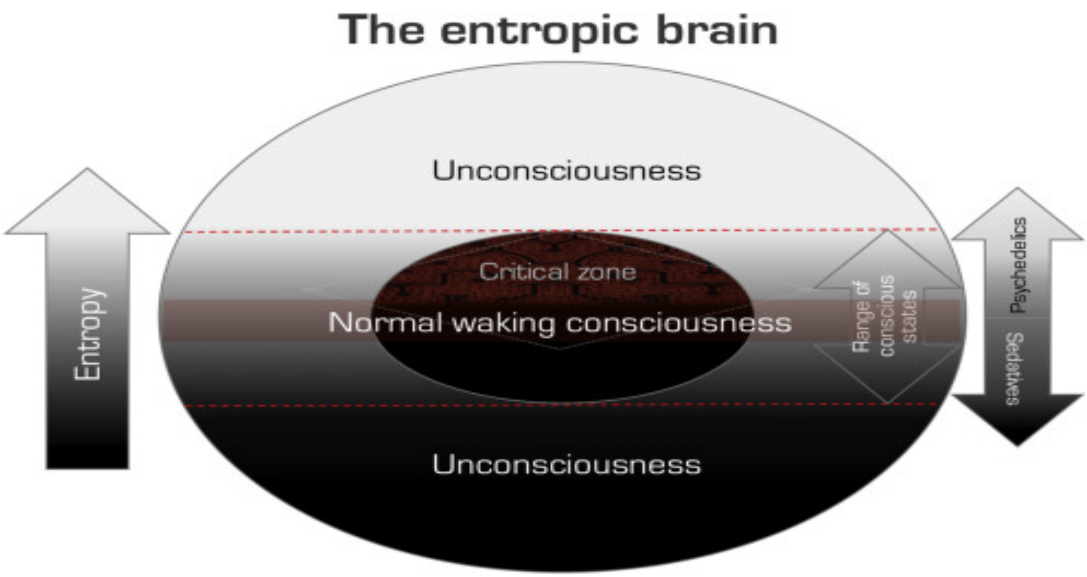

Figure 2. A diagram illustrating the entropic brain hypothesis [8]

In conclusion, the key factor of digital immortality is the preservation of personal identity and consciousness. It's also the most important criterion for judging the success of the approaches.

A great amount of research has been done on digital immortality, and I will discuss four major approaches and my approach to digital immortality. The following approaches are listed from the least plausible to the most.

Approach 1: Reconstruct from the behavior approach

The first approach is the reconstruction from the behavior approach, "in which information about the behavior of an individual is collected and later used to parametrize a generic substrate in order to in some sense reconstruct the individual" [9]. Proponents of this approach include Bainbridge and Rothblatt. It is essentially using data recorded from daily life, including social media posts, legal information, and descriptions from others. I view this as similar to machine learning for AI, which is giving AI a desired target and putting it into a "black box" to train, involving the usage of a neural network. A detailed explanation of the neural network can be found in Appendix A.

For this reconstruction from the behavior approach, it's great if you want to create a person that perfectly mimics you. They will have the exact same reaction as you when facing different situations. Still, their thinking logic, or the "hidden layers", will be completely different than yours. For example, the reason why you want to eat a spicy meal is because it can make you have a better appetite. However, the reconstructed you may want to eat a spicy meal because they enjoy the pain. In others' opinion, you may still be you because the reconstructed you will do the

exact same thing as you. However, your personal identity is lost during reconstruction from behavior since it fails to reach the pasfurconsistency. Also, sometimes it's impossible to produce the original "functions" just from the outcome. (e.g., a linear transformation with a non-zero kernel, you will not be able to find an inverse for that transformation). So, the reconstruction from behavior is not plausible; the probability of success is even less than trying to grow an exact copy of your brain with your DNA.

Approach 2: Reconstruction from a scan approach

The second approach is the reconstruction from a scan approach. "This approach may involve plastination or cryonic suspension, perhaps through a controlled euthanasia, with a view to detailed scanning at a later point" [9-10]. To my understanding, this approach is mainly to collect information, including structural and functional information, through scans. And then trying to reconstruct an artificial brain through microengineering.

A surgery is required for this approach. The person's state, including proteins and lipids, will be locked by chemical fixatives like glutaraldehyde and osmium tetroxide. Then, the brain will be stained with heavy metal for better scanning. Water will be replaced by pasic resin to keep the tissues in shape. A low-temperature sharp diamond blade will be used to slice brain tissue is micro-level stripes. The electrscopy will emmit electron beams to scan the top few nano meter of the strip and then the top layer will be removed by lazer. After thousands of repeats, a complete brain scan (connectome) will be stored in a computer. Then, an artificial brain can be reconstructed from the scan.

Although this process sounds very reasonable, it has a big flaw: your personal identity's existence is not continuous throughout the process. You will be dead at the end of step three because there will be no brain activity after those toxins (chemical fixatives) are added, which

equals brain death. Also, no matter how advanced those scanning machines are, there will still be some inaccuracies that keep the reconstructed copy from being exactly the same as you. Even if, sometime in the future, we have a perfect imaging machine that can make an exact copy, multiple copies can be made, and all of them can claim to be you. It's unreasonable to claim that all of them are you and that they are all separate individuals, proving that they are just merely a copy of you. So that's why I judge the reconstruction from a scan as not plausible because the only thing it can do is making a copy of you.

Approach 3: Instant replacement approach

The third approach is the instant replacement approach. Instant replacement is very similar to the reconstruction from a scan approach. It shares some flaws as the reconstruction from a scan approach. Nonetheless, an instant replacement will solve the problem of multiple copies because it is a replacement process, and there will be only one individual being produced. The instant replacement approach requires a non-destructive scan, which can keep you alive during the scanning. X-ray holographic nanotomography is a great candidate for the non-destructive scan. The detailed information of X-ray holographic nanotomography can be found in Appendix B. With X-ray holographic nanotomography, we can know the cellular composition and arrangement of a neural circuit, which helps us reconstruct the digital version of those neural circuits. With a large field of view, X-ray holographic nanotomography can provide us with a detailed framework of how information flows in the brain. Although today's X-ray holographic nanotomography does not have the resolution in order for us to reconstruct in molecule scale[6], it is unnecessary for us to reconstruct every tiny part of our brain with exactly the same structure, which is also almost impossible (unless we have the technology to

[6] The sizes of molecules generally range from 0.1 nm to 1 nm.

reconstruct in the plank length[7] scale). What makes us unique is not the matter and energy we contain but the information we process. So, as long as we can reconstruct the brain that can contain our distinct information, this method of instant replacement can be considered successful. Some might argue that those tiny structures also contain pieces of information. However, those pieces of information are generic, just like the composition of a carbon-12 atom is 6 protons, 6 neutrons, and 6 electrons, which does not need to be replicated for every individual. In addition, this relatively low resolution is caused by optics focusing, stage movement precision, and reconstruction algorithm performance and is not the fundamental physical limit for the X-ray holographic nanotomography, which can reach the sub-nanometer. So, with X-ray holographic nanotomography, a complete 3-D connectome of your brain can be generated with high resolution. With this connectome, a brain that is identical to yours can be created. The new artificial brain will be briefly connected to your biological brain and activated by your biological energy and action potentials in your brain. After the activation, your biological brain will be destroyed, and you will be alive in the artificial brain.

The major flaw of the method is that there are too many changes happening in a short time interval. (100% of your brain components changed within one day) Borrow the concept of the Ship of Theseus[8]. People might hesitate to choose if the change lasts for a long time. However, if the replacement is done within a day, most people will deny that the ship is the old one.

---

[7] A plank length is equal to 1.6e-35 meter.

[8] In Greek mythology, Theseus, the mythical king of the city of Athens, rescued the children of Athens from King Minos after slaying the minotaur and then escaped onto a ship going to Delos. Each year, the Athenians would commemorate this by taking the ship on a pilgrimage to Delos to honour Apollo. A question was raised by ancient philosophers: If no pieces of the original made up the current ship was it still the Ship of Theseus? Furthermore, if it was no longer the same, when had it ceased existing as the original ship?

Approach 4: Gradual replacement approach

The fourth approach is the gradual replacement approach, which is to surgically replace the nervous system part by part. "The synthetic substrate is assembled alongside the pre-existing biological substrate. The biological substrate may be gradually disassembled, but they operate in parallel for a time to implement the same individual." [10] The first part of this process is the same as the instant replacement approach. It still needs a non-destructive scan to build a template for the replacement. Instead of instant replacement of the whole brain, the replacement is done piecewise. The brain is reconstructed, part by part. The parts are extremely small, just like "dx" in calculus. As soon as a brain part is reconstructed, the corresponding part of the biological brain will be carved out, and the artificial part will be connected to you. To connect biological neurons and artificial neurons, sub-nanorobots are needed for nano surgery. One challenge is that after replacing the outer area of the brain, how are we going to replace the inner parts without exerting much impact on the rest of the brain? This approach is by far the greatest because it preserves the personal identity: brain activity will never stop during the process. Study shows that "More than 1 million new neural connections are formed every second in the first few years of life." [13] So, as long as the rate of replacement does not exceed 1 million per second, you will still be able to maintain your consciousness. "The human brain consists of 100 billion neurons and over 100 trillion synaptic connections" [14]. The average procedure time for adults should be 100 million seconds, which is less than 3.17 years[9]. Thus, the procedure time for the gradual replacement should be longer than 3.17 years to safely preserve consciousness, personality, and memories, which I will refer to as Asher's Second Rule of Digital Ascension: the speed of neural replacement, modification, and addition should not exceed the speed of

[9] One year are assumed to have 365.25 days due to the existence of leap year

natural neuronal changes in percentage (0.00001%/s). If Asher's First and Second Rule of Digital Ascension are satisfied, following the gradual replacement approach, you, yourself, will become immortal. The only con is that after the process, you will remain vulnerable: you will still die in a plane crash, and AI will still be smarter than you.

Approach 5: Building on top approach

The fifth approach is my approach, which is the building-on-top approach. Instead of replacing the natural brain with neural prostheses (as used in the gradual replacement), an artificial layer will be built on top of it. Just like the neocortex grows on top of the mesocortex and allocortex due to evolution, which allows humans to think, the sicocortex* (the new artificial cortex) can be implanted on top of the neocortex. The sicocortex should be able to process the function of a computer and grant humans the function to perform fast calculations and vast simulations. The building of the sicocortex should be part by part with the time of a natural human lifespan. By the time the natural brain dies, the sicocortex will be able to replace the function of the natural brain, due to the plasticity of the brain. As long as Asher's first rule is satisfied, the pasfurconsistency will be met. The sicocortex does not even have to be physically connected to the neocortex, it can be entirely remote. "The three labs, scattered across Padova, Italy, Zurich, Switzerland, and Southampton, England, collaborated to create a fully self-controlled, hybrid artificial-biological neural network that communicated using biological principles, but over the internet"[20]. A Hybrid artificial-biological neural network means there are both artificial neurons and biological ones. Artificial-to-biological pathways are also established within the network. Microelectrodes that can receive digital signals are physically connected to the biological neuron. By doing so, digital signals can be converted to electrical

signals, just as in biological neurons, chemical signals are converted to electrical signals. Those connections are just like the synaptic cleft, however, these gaps can have a distance of hundreds of kilometers. For the building-on-top approach, we do not need to physically rebuild neurons. We can just put a large number of microelectrodes on the neocortex that connects to the digital sicocortex that can be placed anywhere on Earth. The first time connecting, the sicocortex should be small, maybe just a few neurons. As you spend your daily life learning, experiencing, and interacting, new neurons will grow (at least the growth is stimulated) in the sicocortex. New connections will be formed. New memories will not only form in the hippocampus, basal ganglia, and cerebellum but also form in the sicocortex. Whenever you are thinking, new connections and new neurons will also form in the sicocortex. In the beginning, 100% of you may be composed of your biological brain. As time goes by and your sicocortex becomes larger and larger, when your biological brain dies due to aging, your biological brain may only compose 0.00001% of you. So, the death of your biological brain will have very little impact on you as a whole. Borrowing the analogy of Ship of Theseus, my argument is that is should be the same ship because just like us humans, due to respiration and food consumption, most of atoms in our body will not be the same: "About 98% of the atoms in the human body are renewed each year"[26][10]. This shows that the building-on-top method won't change the person into a new one, even when the majority of the composition is not original. This has even greater implications that you can be the internet and the internet can be you, making you omnipresent because where the internet is, you are as well. You can be physically present on a trip to Virginia, U.S., and your "proxy" can be eating a hotpot in Chengdu, China. Using the building-on-top approach allows us to bypass lots of technique difficulties mentioned before: The high-resolution

[10] This surprising fact is discussed by Dr. Paul C. Aebersold of Oak Ridge in the latest Annual Report of the Smithsonian Institution.[26]

scan required to reconstruct the brain, the sub-nanorobots needed for the nanosurgery, and the accurate slicing and imaging with energy/electron beams.

The reason we do not need the sicocortex be biological or pseudobiological is that the reason why we are specifically ourselves is not because of our molecular composition, but the information (the presence or absence of a type of substance at a certain location and certain time) contained in those configurations. As long as the digital sicocortex has the same information processing process, one's identity can be preserved.

Construct digital brain with computational neuroscience models

The only major problem that we need to solve is how to construct a digital brain, which has a lot to do with computational neuroscience[11]. To construct a digital brain, we need first to construct a neuron.

Due to the electric properties of the neuron membrane, neuron signaling can be mimicked through computer programs. Detailed explanations can be found in Appendix D.

One of the most important properties of a neuron is the firing or the action potential, which is a rapid uprising in the membrane potential that can propagate (down the axon). This is like 0 and 1 in neuron language, where the absence of an action potential means 0, and the presence of an action potential means 1. There are no numbers in between because all the action potentials are uniform in magnitude, no matter what input current the neurons are receiving. A voltage reset is followed by the action potential, caused by the deactivation of the sodium channel and the activation of the potassium channel. So, in the simplest model, the Leaky

[11] A sub branch of neuroscience. *An Introductory Course in Computational Neuroscience* by Paul Miller [21] is a great book to get started in computational neuroscience. It's also where I get lots of understanding in constructing digital neurons from.

Integrate-and-Fire Model, the membrane potential is reset to the steady potential (V_{reset}) as soon as it's greater than the threshold potential (V_{th}).

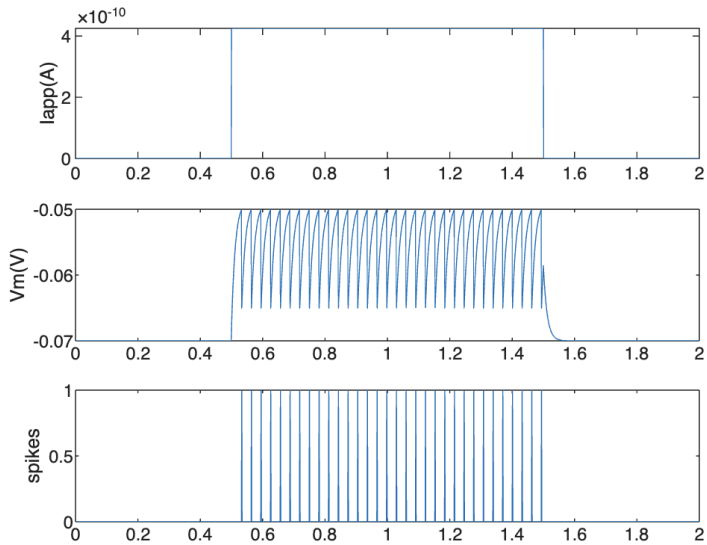

Figure 3. The neuron is modeled by the Leaky Integrate-and-Fire model with $I_{app} = 425pA$ $E_L = -70mV$; $R_m = 50M\Omega$; $G_L = 20nS$ $C_m = 250nF$. *The firing threshold of the neuron is at* $V_{th} = -50mV$ *, and the reset potential is* $V_{reset} = -65mV$.

Generated by code_1_Leaky_Intergrate_and_fire.m.[12]

The Leaky Integrate-and-Fire Model (LIF) is viewed as the simplest model because it generalizes all types of ion channels into one, represented by a single conductance G_L. The change of membrane potential V_m can be represented by a single first-order differential equation:

$$C_m \frac{dV_m}{dt} = G_L(E_L - V_m) + I_{app}$$

And each time the membrane potential reaches the threshold V_{th}, it will be reset to the reset potential V_{reset}. By doing so, the action potential will be incomplete because, in this model,

[12] All my matlab codes are in appendix C.

neurons never have a chance to reach positive membrane potential, where the real neurons will reach. The rate of the membrane potential is a concave-down graph because the smaller the difference between the membrane potential and the steady state V_{ss}, the smaller the membrane potential increment.

The steady state of the membrane potential can be calculated when the change of membrane potential equals zero ($\frac{dV_m}{dt} = 0$). We get:

$$V_{ss} = E_L + I_{app}/G_L$$

This is also the equilibrium the system is trying to reach. So, if the steady state is lower than the threshold, the neuron will not fire[13].

The firing rate is a very important parameter for neurons, which codes the information that neurons try to convey, including the strength of the signals. Since the membrane potential can be expressed as a function over time, the time for a neuron to fire (T) is when $V(T) = V_{th}$:

$$V_{th} = V_m(T) = V_{ss} + (V_{reset} - V_{ss}) * exp(-T/\tau_m) \qquad where\ T > 0\ and\ V_{ss} > V_{th}$$

The interspike interval (ISI) is equal to the time for a neuron to fire (T) and can be calculated by solving the function above:

$$ISI = -\tau_m ln(\frac{V_{ss}-V_{th}}{V_{ss}-V_{reset}}) = \tau_m ln(\frac{V_{ss}-V_{reset}}{V_{ss}-V_{th}})$$

The firing rate will be inverse of ISI and can be expressed as a function over the input current (I_{app}) since $V_{ss} = E_L + I_{app}/G_L$:

$$f(I_{app}) = \frac{1}{\tau_m ln(\frac{V_{ss}-V_{reset}}{V_{ss}-V_{th}})} = \frac{1}{\tau_m ln(\frac{E_L + I_{app}/G_L - V_{reset}}{E_L + I_{app}/G_L - V_{th}})}$$

[13] This always holds true in this Leaky Integrate-and-Fire Model, but may not in more complex models. (e.g. hyperpolarization firing)

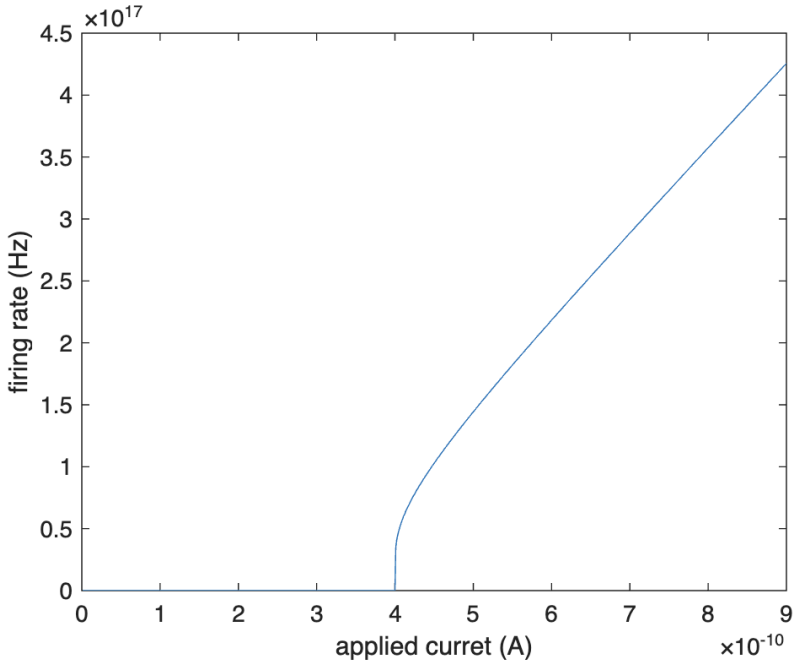

Figure 4. The fire rate of a neuron is plotted against the input current that ranges from 1 nA to 900 nA. Other parameters remain the same as in Figure 3. This graph is generated by code_2_firing_rate.m.

If you are careful, you may notice that the firing rate is very abnormal. 1e17 fire per second is just impossible for a normal neuron[14]! This is because we didn't calculate the refractory period. The refractory period is the minimum time between two neurons firing (like a resting period for neurons). In real neurons, the refractory period is caused by the clamping of the sodium ion channel (the deactivation) and the overshoot caused by the opening of the potassium ion channel.

There are three general methods to simulate the refractory period. The first method is the forced Voltage Clamp method, which tries to mimic the clamp of the sodium ion channel. It works by fixing the membrane potential to the reset potential for the whole refractory period (τ_{ref}). The second method is Refractory Conductance, which tries to mimic the hyperpolarization

[14] average cortical neuron fires around 0.16 times per second [22]

current of the potassium ion channel. It creates a refractory conductance that increases by a constant (ΔG) each time an action potential is fired and decays slowly through time:

$$\frac{dG_{ref}(t)}{dt} = -\frac{G_{ref}(t)}{\tau_{ref}}$$

An additional term is also added to the membrane potential formula, which E_K is the reversal potential for potassium:

$$G_{ref}(t)(E_K - V_m(t))$$

With this method, the force reset step in the LIF model can be dropped. The third method is the Raised Threshold. It prevents neurons from firing by raising the firing threshold (V_{th}), which will no longer be a constant, but a function over time, where $V_{th}^{(0)}$ is the initial threshold value:

$$\frac{dV_{th}(t)}{dt} = \frac{V_{th}^{(0)} - V_{th}(t)}{\tau_{ref}}$$

The value also increased by a constant ΔV each time after a spike.

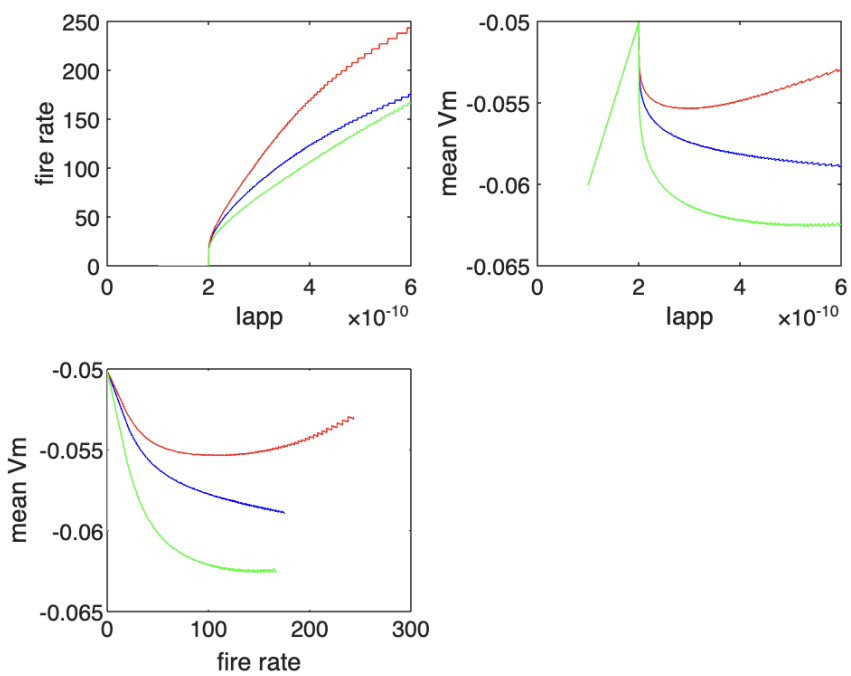

Figure 5. Three different methods are used. The plots are fire rate against time, mean membrane potential against time, and mean membrane potential against fire rate. Blue corresponds to the Forced Voltage Clamp method (method 1), green corresponds to the Refractory Conductance Method (method 2), and red corresponds to the Raised Threshold method (method 3). Those plots are generated by code_3_refractory_period.m.

By using the refractory period, the firing rate is significantly lowered from 1e17 to 1e2.

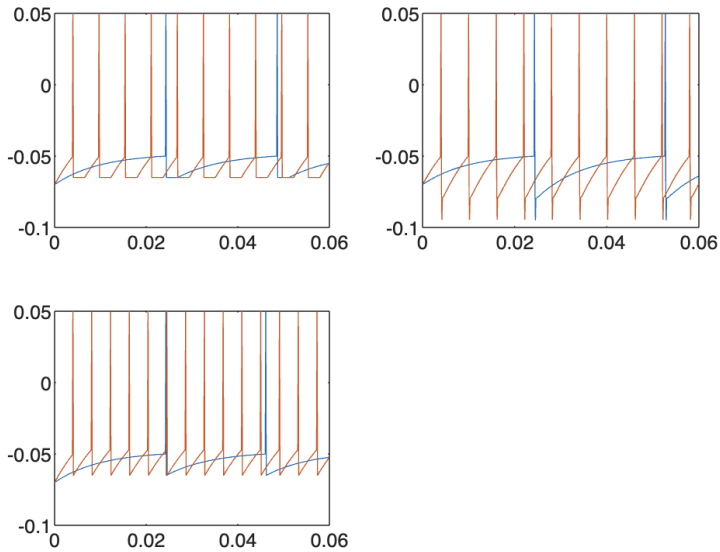

Figure 6. The plot of membrane potential against time with two different input currents. Blue corresponds to an input current of 120 pA. Orange corresponds to an input current of 500 pA. Those plots are also generated by code_3_refractory_period.m.

From those plots, we can see some distinct characteristics of those methods. Method 1 (top left) has a period of absolute reset. Method 2 (top right) has membrane potential overshoot. Method 3 (bottom right) has a higher average firing threshold. Another difference between the

membrane potential plot and the basic LIF plot is the peak, which should be in a real action potential. However, the peaks in those plots are a very rough simulation, achieved by setting membrane potential to the peak voltage (V_{peak}) immediately after crossing the threshold.

So how do we find a better way to simulate the action potential? The Exponential Leaky Integrate-and-Fire (ELIF) model does a better job. In the real cell's action potential plot, an inflection point is observed where the membrane potential turns from concave down to concave up. So, to simulate this, we can add an exponential component to the general membrane potential expression:

$$C_m \frac{dV_m}{dt} = G_L(E_L - V_m + \Delta_{th} exp(\frac{V_m - V_{th}}{\Delta_{th}})) + I_{app}$$

Where Δ_{th} denotes the range of voltage from the threshold where the exponential term starts to take effect. A maximum membrane potential (V_{max}) is also necessary. The membrane potential is reset to V_{reset} as soon as it reaches V_{max}.

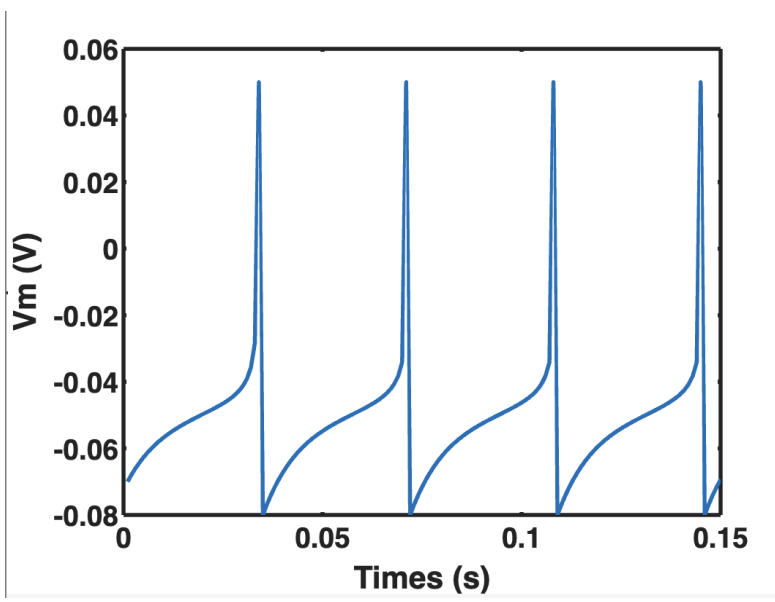

Figure 7. The membrane potential graph against time with an applied current of 210 pA. The figure is generated by code_4_Exponential_Leaky_Intergrate_and_Fire.m.

In this plot, you can clearly see the inflection point and a better peak.

The natural neurons are adaptive, which means that the firing rate is the fastest when receiving stimuli the first time and gradually decreases over time. This is also why when you are drinking something sweet, the first sip is always the sweetest. To address this issue, the Adaptive Exponential Leaky Integrate-and-Fire (AELIF) model is needed, which adds a hyperpolarization spike-rate adaptation current to the general equation. This adaptation current increases by a constant (Δ_{SRA}) each time after a spike, and the rate of decrease is proportional to the magnitude of the current. It also has a conductance that is proportional to the energy potential ($G_{SRA}(V_m - E_L)$). The update of the adaptation current is:

$$\tau_{SRA}\frac{dI_{SRA}}{dt} = G_{SRA}(V_m - E_L) - I_{SRA}$$

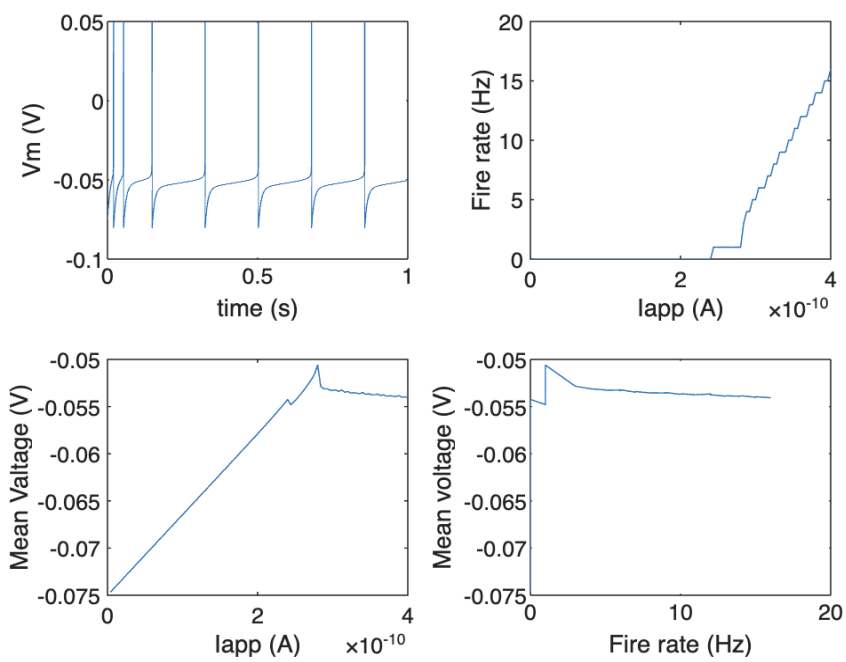

Figure 8. The AELIF neuron's reaction with varying input current (4pA-400pA). Top left is the membrane potential as a function of time with the input current of 400pA. The top right is the

Fire rate as a function of input current. The bottom left is the mean voltage as a function of input current. The bottom right is the mean voltage as a function of fire rate. Produced by code_5_AELIF.m

The AELIF model probably achieved the greatest realistic potential that the LIF model could be. First, it has an adaptive current (see Figure 16 top left) where the firing rate decreases as the stimuli continue over time. Second, the fire rate increases as the input current increases. (see Figure 16 top right). Third, the Mean voltage also increases as the input current increases until the failure around 300pA. (see Figure 16 bottom left). The failure could be overcome by adding more adjustments like a dynamic threshold, but this is enough to illustrate the point. Last, the firing rate is kept well below 20 Hz by the refractory period.

The membrane potential to time graph looks very uniform; the shape of the spikes is the same. There are no fluctuations besides the spikes. The curves are smooth. However, in the real neurons, things are different. The underlying reason is the Brownian movement[15]. In short, Brownian movement is the random movement of particles (suspended in a medium). Which caused lots of random fluctuations in the real neurons' membrane potential. To simulate the randomness in neurons, a Homogeneous Poisson process comes in handy. A Poisson process is a model for a series of discrete events where the average time between events is known, but the exact timing of events is random. The arrival of an event is independent of the event before (waiting time between events is memoryless)[24]. Discrete events are instantaneous events, like neuron firing. The Poisson process is a point process that averages the values of those discrete events. In this paper, the Poisson process will be the chance of neuron firing in a very small time

[15] In Feynman Lectures on Physics Vol. I Ch. 41, there are more detailed explanations on the Brownian movement. Great physics book to read. [23]

interval(δt). The homogeneous part means that this Poisson process has a proportionality constant (r) that is independent of time (t). So the probability of a neuron firing at a small interval is $r\delta t$. Using this, the formula of the probability of N spikes in a time interval T can be derived:

$$P(N) = \frac{(rT)^N e^{-rT}}{N!}$$

So why is this formula important? With this, we can calculate the interspike interval (ISI). The ISI is simply a period (T) of no spike followed by a spike in a very short interval (δt), which is:

$$P(0)r\delta t = e^{-rT} r\delta t$$

With a tiny time bin (δt), the chance of multiple firing can be ignored. ($P(N > 1) \approx 0$). So the chance of silence is $P(0) = 1 - p(1) = 1 - r\delta t$. This allows us to generate a spike train ($ST(\delta t)$) with Gaussian distribution[16]. The spike train only contains 0 and 1, where 1 represents the presence of a spike in the short time interval and 0 represents the absence of the spike. We also need a function that can generate a random integer ($R(x)$), with the result independent from the input. With all that, we can get the function:

$$ST(\delta t) = \frac{e^{\frac{1}{2\pi\sigma} e^{-\frac{R(\delta t)^2}{2\sigma^2}}}}{1+e^{\frac{1}{2\pi\sigma} e^{-\frac{R(\delta t)^2}{2\sigma^2}}}} < r\delta t$$

Where σ is the standard deviation we observed.

We can also add random components in the input current that causes changes in the membrane potential, which has a standard deviation of $\sigma_I/\sqrt{\Delta t}$. (σ_I is the normal standard deviation of the current (I) with the unit of $pA \cdot s^{0.5}$, Δt is the time step we choose in the simulation). The applied current (I_{app}) will be the base current (I_0) plus the random term:

[16] It's also called Normal distribution, or belt curve.

$$I_{app}(\Delta t) = I_0 + \frac{e^{\frac{1}{2\pi\sigma}e^{-\frac{R(\Delta t)^2}{2\sigma^2}}}}{1+e^{\frac{1}{2\pi\sigma}e^{-\frac{R(\Delta t)^2}{2\sigma^2}}}}\sigma_I/\sqrt{\Delta t}$$

With the random fluctuation (noise) added to the AELIF model, the membrane potential graph will be closer to the one in vivo.

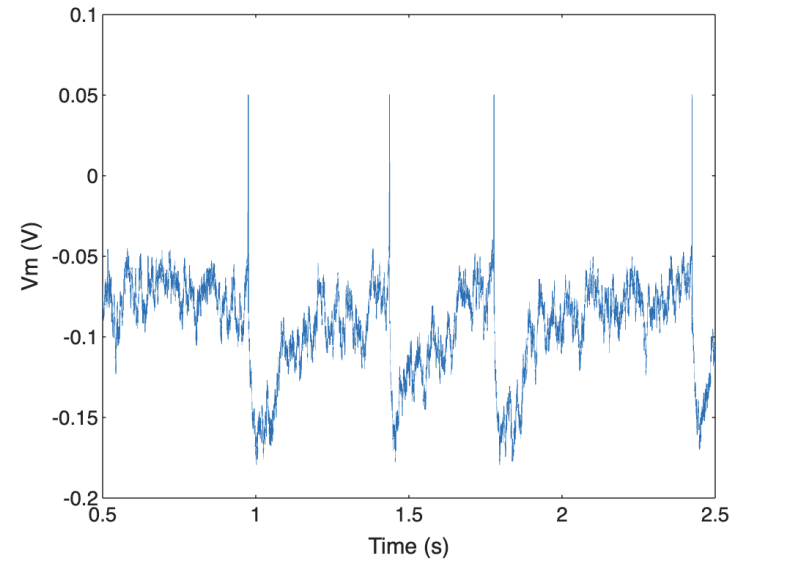

Figure 9. The membrane potential graph. Generated by code_6_AELIF_with_noise.m

The fluctuations occur in the membrane potential graph, which helps to keep the interspike interval from being a constant. This allows us to acquire valuable neuron properties from the simulation. For example, the coefficient of variation (CV)[17] of the ISI can be calculated, which is closely related to the firing rate of a neuron. It can be calculated by dividing the standard deviation by the mean of a set of ISIs. The Fano Factor is also a measure of variance, calculated by dividing the variance by the mean.

[17] Spike trains of cortical neurons measured in weakful mammals typically have CVs near onr or greater than one. [21]

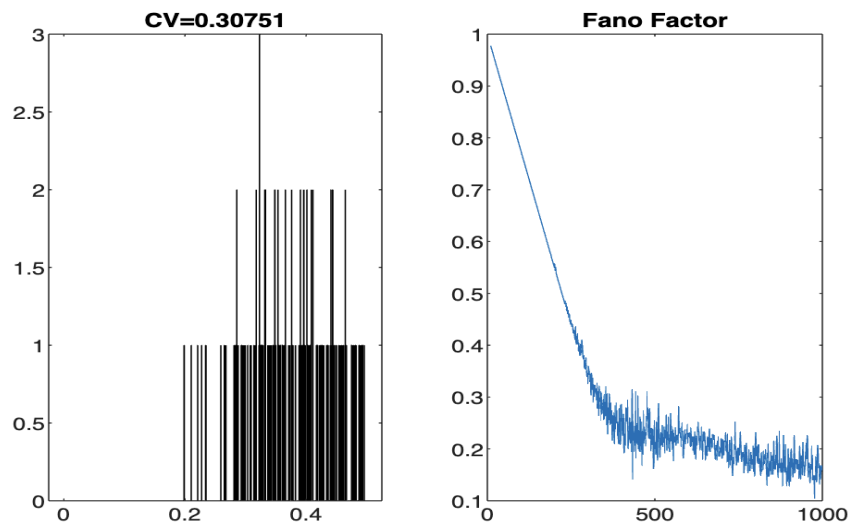

Figure 10. Left is a histogram of the ISIs with the bin of $25\Delta t$ (0.25ms). Right is the Fano Factor plotted against the time bin width. Generated by code_6_AELIF_with_noise.m

We can describe the histogram (Figure 18 Left) as roughly symmetric, which shows that the impact of input current to the interspike intervals is linear. It has a mean of $0.35 * 0.25 = 0.0875\, ms = 8.75 \cdot 10^{-5} s$. By taking the inverse, we can get the firing rate is $11429\, Hz$. This shows that I didn't do a good job simulating the refractory period in code 6. (The parameter of the refractory period can be adjusted.) However, this also shows us the usefulness of the analysis of neuron statistics.

With everything added, including refractory period, exponential spike mechanism, spike-rate adaptation, and the noise/random factors, the Leaky Integrate-and-fire reached its best form. However, this model is far from perfect: it doesn't account for the intricate interactions of all type of ion channels, like the conductance change of sodium, potassium and chloride ion channels; the properties of neurons are also not only just dependent on the membrane potential but also others like calcium ion influx; and in the LIF model, the neurons are considered as a

whole with no difference in spatial variations, where as in real neurons, the electrical properties vary from regions (the conductance in dendrites is different than in the axon or the soma).

To solve those problems, a better model is introduced — the Hodgkin-Huxley Model[18], in which the action potentials are generated by the activities of sodium and potassium ion channels. The sodium ion channel can be viewed as a positive feedback loop: activated by the depolarization of membrane potential, allowing the influx of sodium ions that will further depolarize the membrane. On the other hand, the potassium ion channel can be viewed as a negative feedback loop: activated by the depolarization of membrane potential, allowing the outflow of potassium ions that will hyperpolarize the membrane. Sodium ion channels react quickly to the high membrane potential, allowing the peak of the action potential to appear. The sodium ion channel will be inactivated soon after the activation, preventing constant high membrane potential, and the slow activated potassium ion channel will bring the membrane potential down to resting.

The Hodgkin-Huxley Model is based on the membrane potential (V_m) and three gating variables: sodium activation variable (m), sodium inactivation variable (h), and potassium activation variable (n). Those gating variables have a value of between 0 and 1, representing the proportion of ion channels in a particular state. Multiplying those gating variables can show the fraction of opening channels that are able to transmit current.

The dynamics in the Hodgkin-Huxley Model can be described with four differential equations:

$$C_m \frac{dV_m}{dt} = G_L(E_L - V_m) + G_{Na}^{(max)} m^3 h(E_{Na} - V_m) + G_K^{(max)} n^4 (E_k - V_m) + I_{app}$$

[18] In 1963, Alan Hodgkin and Andrew Huxley (jointly with John Eccles) received the Nobel Prize for their work, which began the field of computational neuroscience when computers were barely available.

$$\frac{dm}{dt} = \alpha_m(1 - m) - \beta_m m$$

$$\frac{dh}{dt} = \alpha_h(1 - h) - \beta_h h$$

$$\frac{dn}{dt} = \alpha_n(1 - n) - \beta_n n$$

The α and β in the equations are rate constants for the gating variables that depend on the membrane potential. α is the rate of moving to the particular state (α_m) is the rate of moving towards the activation of the sodium ion channel. β is the rate of moving away. The rate constants can be calculated using empirical data. In the Hodgkin-Huxley model, the rate constants are:

$$\alpha_m = \frac{10^5(-V_m - 0.045)}{e^{(100(-V_m - 0.045))} - 1} \qquad \beta_m = 4000e^{\left(\frac{-V_m - 0.070}{0.018}\right)}$$

$$\alpha_h = 70e^{(50(-V_m - 0.070))} \qquad \beta_h = \frac{10^3}{e^{(100(-V_m - 0.040))} + 1}$$

$$\alpha_n = \frac{10^4(-V_m - 0.060)}{e^{(100(-V_m - 0.060))} - 1} \qquad \beta_n = 125e^{\left(\frac{-V_m - 0.070}{0.08}\right)}$$

Since some denominators can be zero, limits have to be calculated, and L'Hôpital's rule can be used:

$$\lim_{V_m \to -0.045} \alpha_m = 1000$$

$$\lim_{V_m \to -0.06} \alpha_n = 100$$

With those rate constants, the gating variables can be calculated.

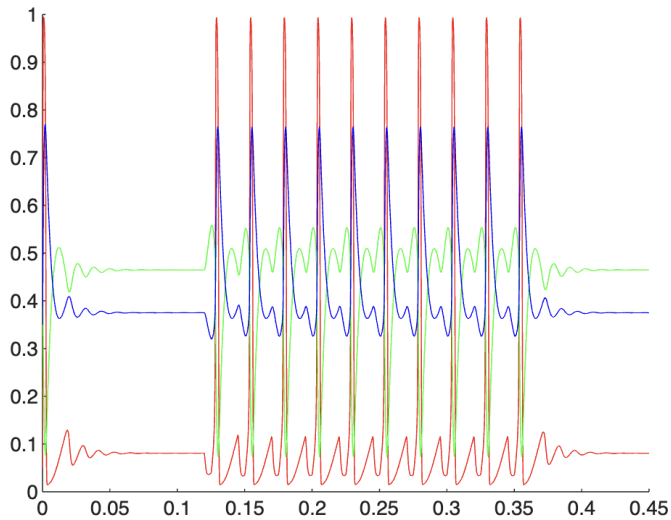

Figure 11. The gating variables are plotted against time. Red represents m, the gating variable for sodium ion channel activation. Green represents h, the gating variable for sodium ion channel inactivation. Blue represents n, the gating variable for potassium ion channel activation. Generated by code_7_Hodgkin_Huxley_Model.m

If you zoom in and carefully observe each peak, you will find the red (sodium activation) is slightly earlier than the blue (potassium activation), providing a window for neurons to spike. The green (sodium inactivation) always has a negative gradient against the red, revealing an inverse relationship (one increase, the other decrease) between two gating variables. The interactions of those gating variables are caused by the change in the membrane potential, while they also have a great influence on the membrane potential.

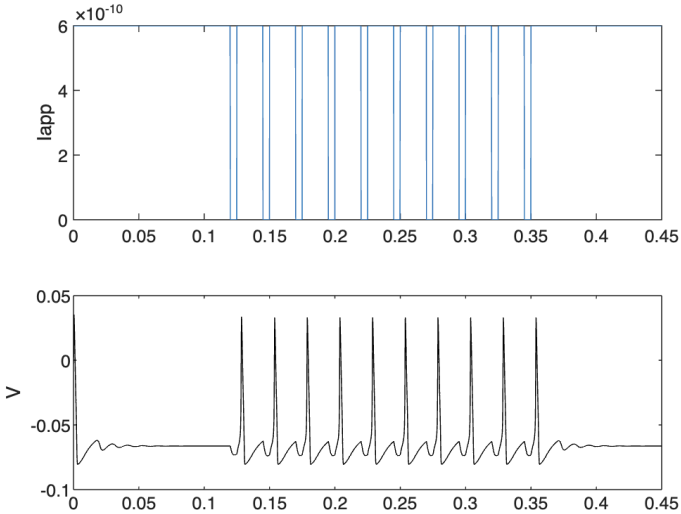

Figure 12. Top: The base input current is 600 pA. There are 10 hyperpolarizing currents. Bottom: The membrane potential is plotted against time. Generated by code_7_Hodgkin_Huxley_Model.m

Some of you might be wondering: Why will those hyperpolarizing currents trigger action potentials? It is because the anode breaks. Hodgkin-Huxley Model models the behavior of Type-II neurons, which are neurons that contain subthreshold oscillations. Just like you pushing or pulling a swing, no matter what direction you push, the magnitude of the swing swinging back and forth will be the same. And if you stop pushing (stop applying current), the swing will stop by itself (membrane potential returns to resting) due to friction (difference between the current membrane potential and the steady state).

Besides this, there are also lots of interesting properties of the Hodgkin-Huxley Model, like bistability, the fittest frequencies of oscillating input current, and the discontinuous jump in firing rate.

The development of the Hodgkin-Huxley Model is based on the giant squid neuron that has a lot of similarities with mammalian neurons. However, one key difference is that neurons in the mammalian cortex are able to produce low firing rate spikes. This is also a key feature of Type-I neurons, whose firing rate increases from zero without any jump. To model this feature, the Connor-Steves model, a variant of the Hodgkin-Huxley model, introduced an additional hyperpolarizing potassium current called the A-current, which has both activation (a) and inactivation (b) gating variables. The membrane potential can be modeled with six differential equations:

$$C_m \frac{dV_m}{dt} = G_L(E_L - V_m) + G_{Na}^{(max)} m^3 h(E_{Na} - V_m) + G_K^{(max)} n^4 (E_k - V_m) + G_A^{(max)} a^3 b(E_A - V_m) + I_{app}$$

$$\frac{dm}{dt} = \alpha_m(1 - m) - \beta_m m$$

$$\frac{dh}{dt} = \alpha_h(1 - h) - \beta_h h$$

$$\frac{dn}{dt} = \alpha_n(1 - n) - \beta_n n$$

$$\frac{da}{dt} = \frac{a_\infty - a}{\tau_a}$$

$$\frac{db}{dt} = \frac{b_\infty - b}{\tau_b}$$

The gating variables of a current do not have the derived rate constant. Still, the voltage-dependent steady state (a_∞, b_∞) and time constant (τ_a , τ_b) can be used to calculate the gating variables. The reversal potential of the A-current (E_A) is very close to the reversal potential of potassium (E_K). The way of calculating those gating variables is given as the following:

$$\alpha_m = \frac{3.8\times10^5(V_m+0.0297)}{1-e^{-100(V_m+0.0297)}} \qquad \beta_m = 1.52\times10^4 e^{-55.6(V_m+0.0547)}$$

$$\alpha_h = 266e^{-50(V_m+0.048)} \qquad \beta_h = \frac{3800}{e^{-100(V_m+0.018)}+1}$$

$$\alpha_n = \frac{2\times10^4(V_m+0.0457)}{1-e^{-100(V_m+0.0457)}} \qquad \beta_n = 250e^{-12.5(V_m+0.0557)}$$

$$\tau_a = 3.632\times10^{-4} + \frac{1.158\times10^{-3}}{1+e^{49.7(V_m+0.05596)}} \qquad a_\infty = \left(\frac{0.0761e^{31.4(V_m+0.09422)}}{1+e^{34.6(V_m+0.00117)}}\right)^{1/3}$$

$$\tau_b = 1.24\times10^{-3} + \frac{2.678\times10^{-3}}{1+e^{62.4(V_m+0.050)}} \qquad b_\infty = \left(\frac{1}{1+e^{68.8(V_m+0.0533)}}\right)^4$$

$$\lim_{V_m\to-0.0297} \alpha_m = 3.8\times10^3$$

$$\lim_{V_m\to-0.0457} \alpha_n = 200$$

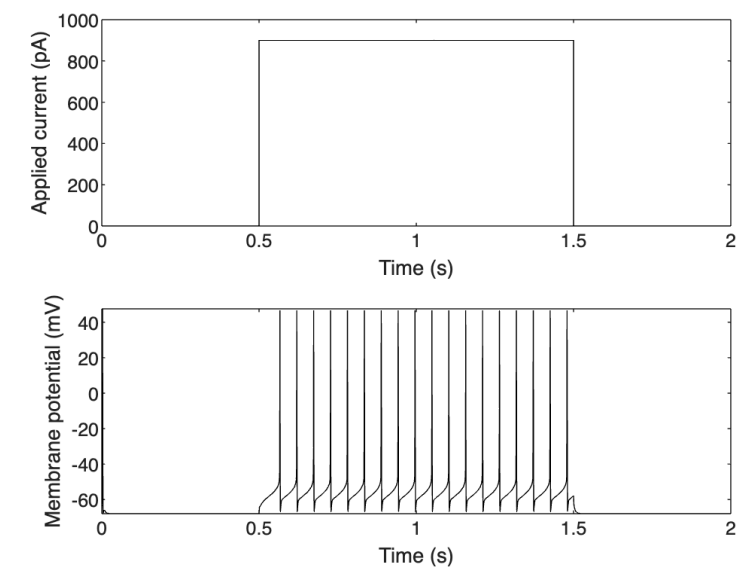

Figure 13. Top: The base input current is 900 pA from 0.5s to 1.5s. Bottom: The membrane potential is plotted against time. If you are careful, you might notice that compared to Figure 20,

there are no oscillations of the membrane potential, showing the difference between the Type II neuron and the Type I neuron. The graph is generated by code_8_Connor_Stevens_Model.

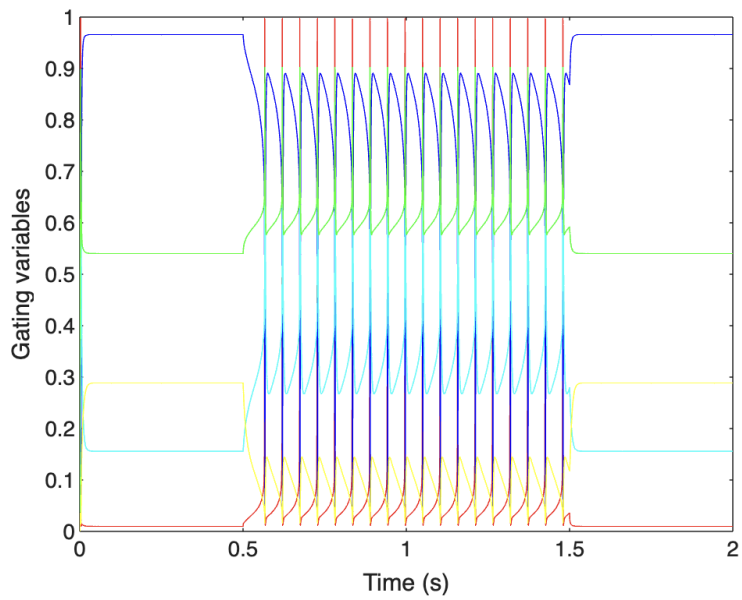

Figure 14. Gating variables are plotted as a function of time. Red is the gating variable for sodium activation. Blue is the gating variable for sodium inactivation. Cyan is the gating variable for potassium activation. Green is the gating variable for A-current potassium activation. Yellow is the gating variable for A-current potassium inactivation. The red is always earlier than everyone else for each spike. A current will also spike with the action potential, preventing the membrane potential from oscillating. This figure is produced by code_8_Connor_Stevens_Model.

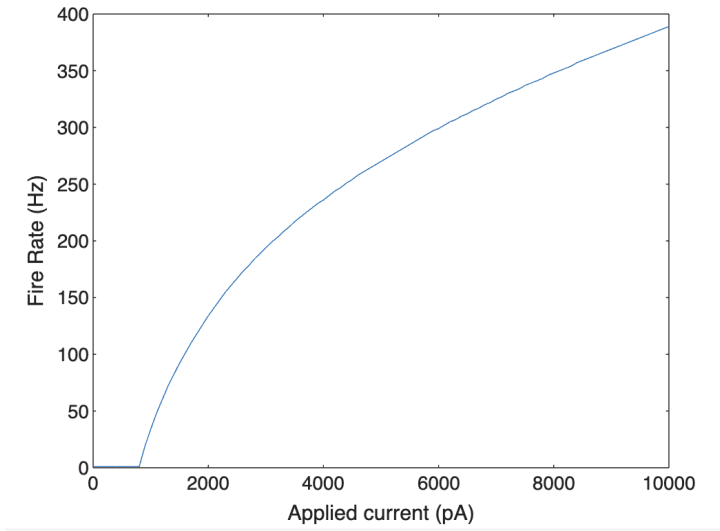

Figure 15. Fire rate plotted as a function of current. The fire rate increases as the magnitude of the applied current increases. Unlike Type-II neurons, there is no jump in the firing rate for Type I neurons. This figure is produced by code_8_Connor_Stevens_Model.

Besides sodium and potassium, calcium also plays an important role in neuron firing, which is mostly influential in thalamocortical neurons. Calcium has a higher concentration in extracellular fluid than in intracellular fluid, partially because the condensation of calcium salt is harmful to the cells/neurons. Thus, the opening of calcium channels, which is known as the T-type calcium channel, will bring a strong depolarizing current to the neuron. The calcium dynamic is slower than that of sodium, generally lasting hundreds of milliseconds. Like sodium channels, calcium channels also have an activation and an inactivation gating variables (m_T , h_T). The calcium dynamics will provide an interesting property—post-inhibitory rebound—for the neurons, producing a positive membrane potential rebond after a period of inhibition, which is quite similar to the anode break. The T-type calcium channels are normally kept inactivated, but when the membrane potential is below the resting membrane potential, the channels will be

deinactivated, producing an inflow of positive calcium ions. The name of this model is very self-explanatory: the thalamic rebound model. The membrane potential can be generated by four differential equations:

$$C_m \frac{dV_m}{dt} = G_L(E_L - V_m) + G_{Na}^{(max)} m^3 h(E_{Na} - V_m) + G_K^{(max)} n^4 (E_k - V_m)$$

$$+ G_T^{(max)} m_T{}^2 h_T (E_{Ca} - V_m) + I_{app}$$

$$\frac{dh}{dt} = \alpha_h(1 - h) - \beta_h h$$

$$\frac{dn}{dt} = \alpha_n(1 - n) - \beta_n n$$

$$\frac{dh_T}{dt} = \frac{h_{T,\infty} - h_T}{\tau_{hT}}$$

Notably, there are no differential equations for sodium activation (m) and calcium activation (m_T). This is because their time constant is 0 because they practically respond spontaneously towards changes in the membrane potential, causing them to possess the same value as the steady state. The voltage-dependent variables are calculated as follows:

$$\alpha_m = \frac{10^5(V_m + 0.035)}{1 - e^{-100(V_m + 0.035)}} \qquad \beta_m = 4000e^{-(V_m + 0.06)/0.018} \qquad \tau_m = 0$$

$$\alpha_h = 350e^{-50(V_m + 0.058)} \qquad \beta_h = \frac{5000}{e^{-100(V_m + 0.028)} + 1}$$

$$\alpha_n = \frac{5 \times 10^4(V_m + 0.034)}{1 - e^{-100(V_m + 0.034)}} \qquad \beta_n = 625e^{-12.5(V_m + 0.044)}$$

$$\tau_{mT} = 0 \qquad m_{T,\infty} = \frac{1}{1 + e^{-\frac{V_m + 0.052}{0.0074}}}$$

$$\tau_{hT} = 0.001e^{15(V_m+0.467)} \qquad if\ V_m < -0.08$$

$$\tau_{hT} = 0.028 + 0.001e^{\frac{-(V_m+0.022)}{0.0105}} \qquad if\ V_m \geq -0.08$$

$$\lim_{V_m \to -0.035} \alpha_m = 1000$$

$$\lim_{V_m \to -0.034} \alpha_n = 500$$

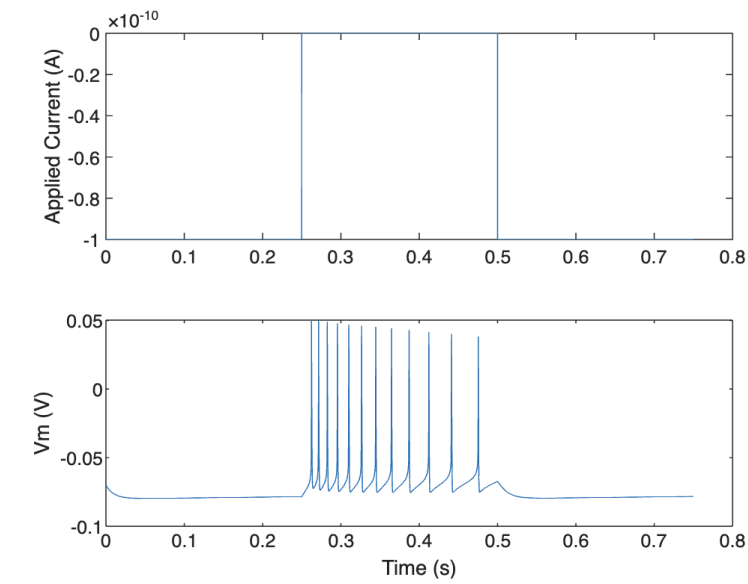

Figure 16. Top: input current plotted as a function of time. The baseline input is -100 pA and has a stimulation input of 0 pA. (equivalent to the stop of the hyperpolarizing current) Bottom: The membrane potential is plotted as a function of time. In this plot, we can clearly see the post-inhibitory rebound. The calcium ion channels are deactivated, producing a long-lasting depolarizing current. As a result, when the hyperpolarizing current stops, decreasing firing is caused by sodium ion channel activation. This figure is generated by code_9_Thalamic_Rebound_Model.m

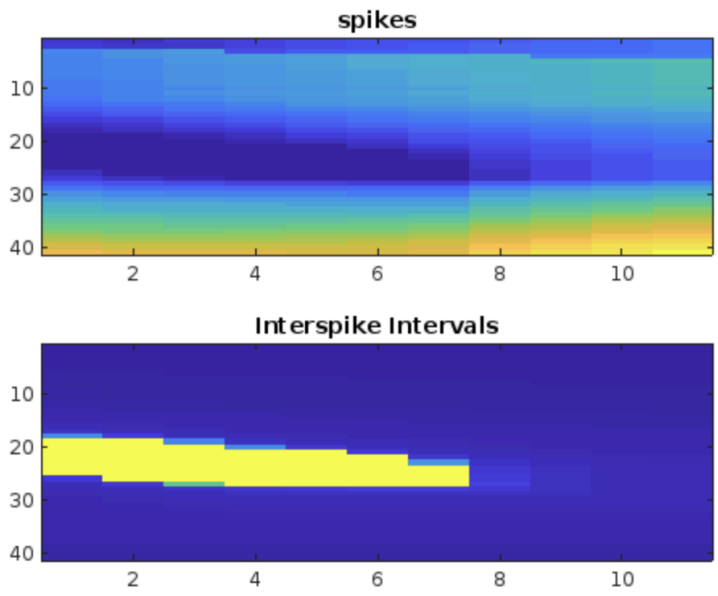

Figure 17. On the y-axis is the baseline current, ranging from -200 pA to 200 pA with a step of 10 pA. On the x-axis is the burst current, ranging from 0 pA to 100 pA with a step of 10 pA. Top: The distribution of spikes. The brighter the color, the higher the value of the block will be. Bottom: The minimum interspike interval of each trial. The inner bright region actually has a value of 100000 dt, meaning that no spikes occurred under that specific condition. The reason why there are spikes with the negative current is because of hyperpolarizing rebound. This figure is generated by code_9_Thalamic_Rebound_Model.m

So far, we have been treating neurons as a whole, uniform existence. However, due to the distribution of different ion channels, different parts of a neuron can be very different, especially for the dendrites and the soma. In real neurons, different parts of neurons have different functions. Generally speaking, dendrites are like receptors, receiving signals from the outside world; soma is like a central processor, processing signals from dendrites; and the axon is like a transmitter, sending information processed by soma to other neurons. The Pinsky-Rinzel model,

a member of multi-compartmental models, takes those differences into account, dividing the neuron into dendrite and soma. In this model, soma and dendrite will have different parameters and will be modeled by different differential equations:

$$C_S \frac{dV_S}{dt} = G_L^{(S)}(E_L - V_S) + G_{Na}^{(max)} m^2 h(E_{Na} - V_S) + G_K^{(max)} n^2 (E_k - V_S) + G_{Link}(V_D - V_S) + I_{app}^{(S)}$$

$$C_D \frac{dV_D}{dt} = G_L^{(D)}(E_L - V_D) + G_{Ca}^{(max)} m_{Ca}{}^2 (E_{Ca} - V_D) + G_{KCa}^{(max)} m_{KCa} X (E_k - V_D)$$

$$+ G_{KAHP}^{(max)} m_{KAHP}(E_K - V_D) - G_{Link}(V_D - V_S) + I_{app}^{(D)}$$

Where S is for parameters corresponding to Soma and D is for parameters corresponding to Dendrites. G_{link} correspond to the conductance of connections between the soma and the dendrite. G_{Ca} correspond to the dendritic inward calcium current, contributing to the generation of the calcium spike. G_{KCa} correspond to the outward calcium-dependent potassium conductance, potentially ending the detritic spike. X is a special gating variable that is dependent on the calcium concentration the is related to KCa. G_{KAHP} correspond to the outward after-hyperpolarization conductance, determining the intervals of dendritic spikes. In addition to those two dynamics above, the Pinsky-Rinzel model also includes the dynamics of the calcium concentration:

$$\frac{d[Ca]}{dt} = -\frac{[Ca]}{\tau_{Ca}} + kG_{Ca}^{(max)} m_{Ca}{}^2 (E_{Ca} - V_D)$$

Where k has a unit of M/C (Molar/liter over charge).

The gating variables for the Pinsky-Rinzel model can be calculated as follows:

$$\alpha_m = \frac{320\times10^3(V_s+0.0469)}{1-e^{-250(V_s+0.0469)}} \qquad \beta_m = \frac{280\times10^3(V_s+0.0199)}{e^{200(V_s+0.0199)}-1}$$

$$\alpha_h = 128e^{\frac{-(V_s+0.043)}{0.018}} \qquad \beta_h = \frac{4000}{e^{-200(V_s+0.02)}+1}$$

$$\alpha_n = \frac{16\times10^3(V_s+0.0249)}{1-e^{-200(V_s+0.0249)}} \qquad \beta_n = 250e^{-25(V_s+0.04)}$$

$$\alpha_{mCa} = \frac{1600}{1+e^{-72(V_D-0.005)}} \qquad \beta_{mCa} = \frac{2\times10^4(V_D+0.0089)}{e^{200(V_D+0.0089)}-1}$$

$$If\ V_D > -0.01: \quad \alpha_{mKCa} = 2000e^{\frac{-(V_D+0.0535)}{0.027}} \qquad \beta_{mKCa} = 0$$

$$If\ V_D \leq -0.01: \quad \alpha_{mKCa} = \frac{e^{\frac{V_D+0.05}{0.011}-\frac{V_D+0.0535}{0.027}}}{0.018975} \qquad \beta_{mKCa} = 2000e^{\frac{-(V_D+0.0535)}{0.027}} - \frac{e^{\frac{V_D+0.05}{0.011}-\frac{V_D+0.0535}{0.027}}}{0.018975}$$

$$X = min(4000[Ca], 1)$$

$$\alpha_{mKAHP} = min(20, 20000[Ca]) \qquad \beta_{mKAHP} = 4$$

$$\lim_{V_s\to-0.0469} \alpha_m = 1280 \qquad \lim_{V_s\to-0.0199} \beta_m = 1400$$

$$\lim_{V_s\to-0.0249} \alpha_n = 80 \qquad \lim_{V_D\to-0.034} \beta_{mCa} = 100$$

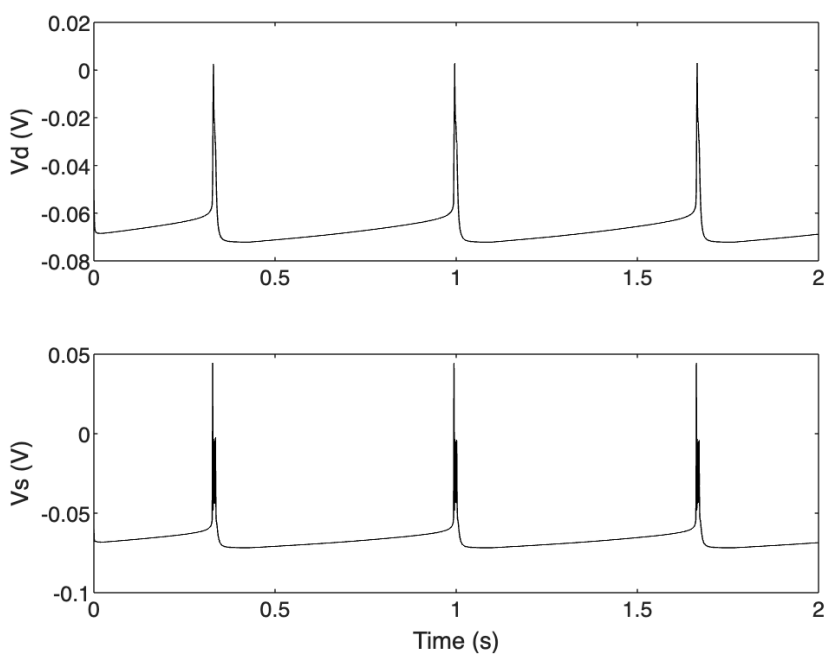

Figure 18. The membrane potential over the whole simulation is plotted against time. The top is dendritic potential, and the bottom is somatic potential. Notably, there is no input current, and the neuron is firing on itself, revealing the intrinsic firing properties of neurons. This graph is generated by code_10_Pinsky_Rinzel_Model.m

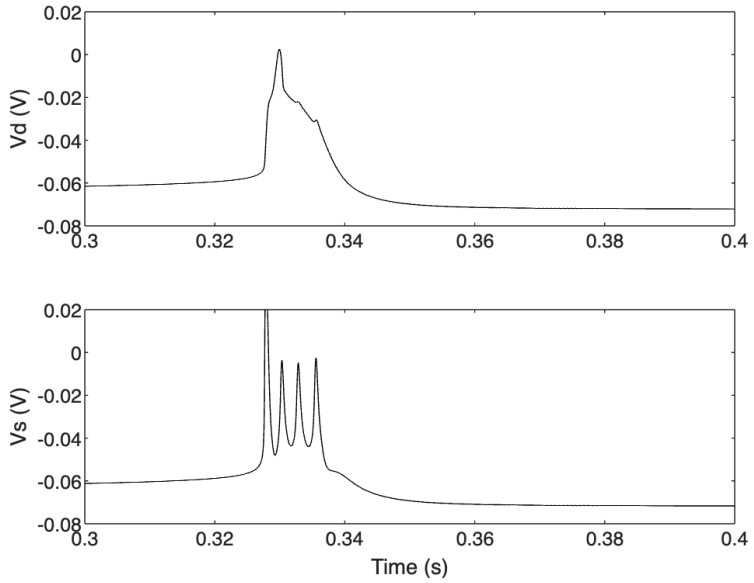

Figure 19. The enlarged graph for the membrane potential plot. In soma, four sodium spikes are triggered in soma because sodium conductance changes with celerity. However, only one calcium spike is triggered in the dendrite because calcium conductance reacts slowly in response to the membrane potential. This graph is generated by code_10_Pinsky_Rinzel_Model.m

I will end the neuron modeling in this paper with the Pinsky Rinzel Model because this model is already well enough for the purpose of this paper. This is not saying that this model is sufficient for the purpose of digital ascension, which requires my further research in college, but it is enough for readers of this paper to understand and be convinced of the approach. I acknowledge the fact that I can’t provide a better model due to my lack of knowledge and

inability to do experiments as a high school student. However, the models I provided above, though reading and learning is enough to provide insight to model a neuron. After modeling a single neuron, the next step is to simulate the synapses and interactions in neurons.

The majority of synapses in mammals are chemical. Unlike electrical synapses with gap junctions, chemical synapses converts electrical signals (action potentials triggered calcium influx causing the vesicle release) to chemical signals (neurotransmitter in the synaptic cleft), and then to electrical signals in another neuron (receptors binds to the neurotransmitters generating current based on receptor type—ionotropic or metabotropic) in the timescale of milliseconds. Synapses are divided into two groups: excitatory synapses that cause depolarization in the postsynaptic neuron and inhibitory synapses that cause hyperpolarization in the postsynaptic neuron. According to Dale's principle: " a given neuron contains and releases only one neurotransmitter and exerts the same functional effects at all of its termination sites"[28], neurons in a circuit can be classified as either excitatory or inhibitory with few exceptions (excitatory D1 and inhibitory D2 receptors for dopamine), releasing only one type of neurotransmitter that can be excitatory or inhibitory. The dominant excitatory neurotransmitter in mammals is glutamate, which causes the inflow of cations, and the dominant inhibitory neurotransmitter is GABA, which causes the inflow of chloride.

To model the synapse, we can generalize the effect of the presynaptic neuron as a current:

$$I_{syn}(t) = G_{syn}(t)(V_{syn} - V_m(t))$$

Where V_{syn} is the reversal potential of the synapse and $G_{syn}(t)$ is the synaptic conductance.

To calculate $G_{syn}(t)$, the simplest model is to use a step increase caused by neuron firing followed by an exponential decay. The dynamic equations are given as follows:

$$\frac{dG_{syn}(t)}{dt} = \frac{-G_{syn}(t)}{\tau_{syn}} + \frac{d\Delta G_{syn}(t)}{dt}$$

$$\Delta G_{syn}(t) = \frac{\Delta G}{\left(\frac{\tau_{decay}}{\tau_{rise}+\tau_{decay}}\right)\left(\frac{\tau_{rise}}{\tau_{rise}+\tau_{decay}}\right)^{\frac{\tau_{rise}}{\tau_{decay}}}} e^{\frac{-(t-t_{spike})}{\tau_{decay}}} \left(1 - e^{\frac{-(t-t_{spike})}{\tau_{rise}}}\right)$$

Where t_{spike} is the spike time, τ_{rise} is the rise time, and τ_{decay} is the decay time that $\tau_{decay} > \tau_{rise}$.

However, in many neurons, the channel opening is due to the graded release of neurotransmitters that has a conductance variable ($G_{syn}(t)$) with a voltage dependent steady state ($G_{\infty}(V_{pre})$) and time constant (τ_{syn}), where V_{pre} stand for presynaptic potential:

$$\frac{dG_{syn}(t)}{dt} = \frac{G_{\infty}(V_{pre}) - G_{syn}(t)}{\tau_{syn}(V_{pre})}$$

$$G_{\infty}(V_{pre}) = \frac{G^{(max)}}{1+e^{\frac{-(V_{pre}-V_{th})}{V_{range}}}}$$

$$\tau_{syn}(V_{pre}) = \tau_{rise} + \frac{\tau_{decay}-\tau_{rise}}{1+e^{\frac{(V_{pre}-V_{th})}{V_{range}}}}$$

$G^{(max)}$ is the maximum conductance and V_{range} is the range of variance of presynaptic membrane potential around the firing threshold V_{th}.

Still, in real neurons, change of synaptic conductance is not only related to presynaptic membrane potential, but also to the firing history. One famous saying in neuroscience is the hebbian rule: “neurons that fire together, wire together”, meaning: “any two cells or systems of cells that are repeatedly active at the same time will tend to become ‘associated,’ so that activity in one facilitates activity in the other”[29]. Dynamical synapse (synaptic facilitation) is one contributor to this rule. Dynamical synapse can be concluded into two kinds. The first one is

short-term synaptic depression, which is the decrease in synaptic conductance increment caused by the depletion of neurotransmitter vesicles. The second one is short-term synaptic facilitation, corresponding to the increase in the release probability of remaining vesicles. By combining those two dynamics, real change in synaptic conductance (ΔG) can be calculated by multiplying maximum conductance (G_{max}), initial release probability of vesicles (p_0), the increased probability by short-term synaptic facilitation ($F\ with\ 1 \leq F \leq F_{max}\ where\ F_{max} = \frac{1}{p_0}$), and the fraction of the remaining vesicles ($D\ with\ 0 \leq D \leq 1$):

$$\Delta G = G_{max} p_0 F D$$

$$\frac{dD}{dt} = \frac{1-D}{\tau_D}$$

$$\frac{dF}{dt} = \frac{1-F}{\tau_F}$$

D and F will also receive step changes after each spike:

$$D \rightarrow D - p_0 F D$$

$$F \rightarrow F + f_{fac}(F_{max} - F) \quad where \quad 0 \leq f_{fac} \leq 1$$

In this simulation, the Poisson process is used to generate presynaptic spike trains with a timestep of (dt) and rate of ($r(t)$). The simulation lasts for four seconds. In the first second, the firing rate is 20 Hz. In the second second, the firing rate is 100 Hz. In the third second, the firing rate is 10 Hz. In the fourth second, the firing rate is 50 Hz.

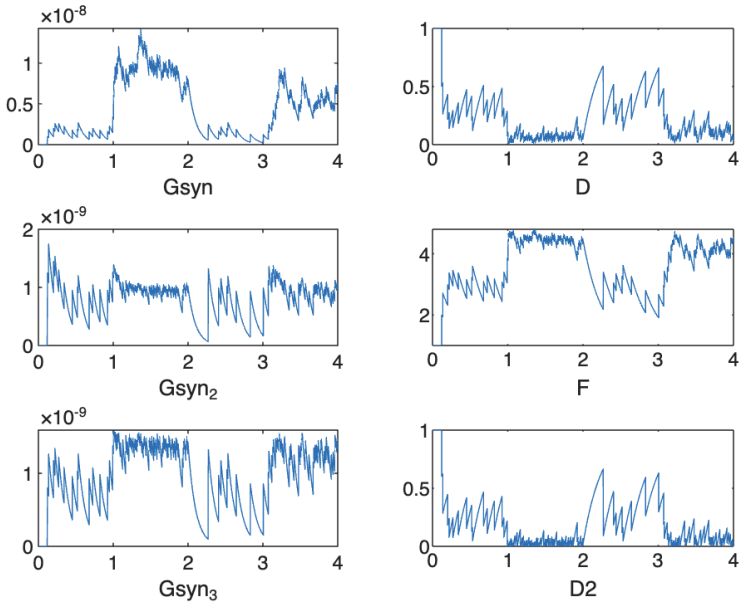

Figure 20. Top left: Plain conductance (conductance without synaptic dynamics) responds to presynaptic spike trains. Top right: Depression factors respond to the spike train without facilitation. Middle left: Conductance response to the spike train with depression alone, where a significant relative decrease in conductance magnitude within 1-2 seconds. Middle right: Facilitation factor responds to the spike train. Bottom Left: Conductance response to the spike train with depression and facilitation. Bottom right: Depression factors respond to the spike train with facilitation. This Graph is generated by code_11_Synaptic_Dynamics.m

A postsynaptic neuron is also generated with the Leaky Integrate-and-Fire model to observe how synaptic dynamics plays a role in neuron signaling.

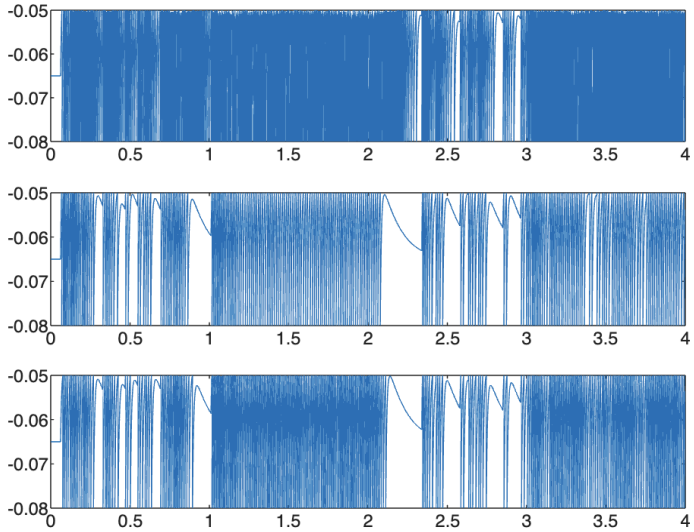

Figure 21. The membrane potential of the postsynaptic neuron plotted against time. Top: Response to plain conductance, where the highest firing rate is observed. Middel: Response to conductance with depression, where the lowest firing rate is observed. Bottom: Response to both depression and facilitation. This graph is generated by code_12_Synaptic_Dynamics_With_A_Postsynaptic_Neuron.m

Using this model, we can simulate how one neuron can influence the other through synaptic connections. Next, we need to simulate the connection of a group of neurons through the connectivity matrix that indicates the strength and direction of the neuron connection. The connectivity matrix is a square non-symmetric matrix with each element having a value between 0 and 1 (just like the matrix in the reconstruct from the behavior approach), while some may just be 0 or 1, indicating the absence or the presence of the connection. The synaptic connection is also directional since the chemical synapses are unidirectional. The cell in i th row and j th column means that the presynaptic neuron i is connected to the postsynaptic neuron j.

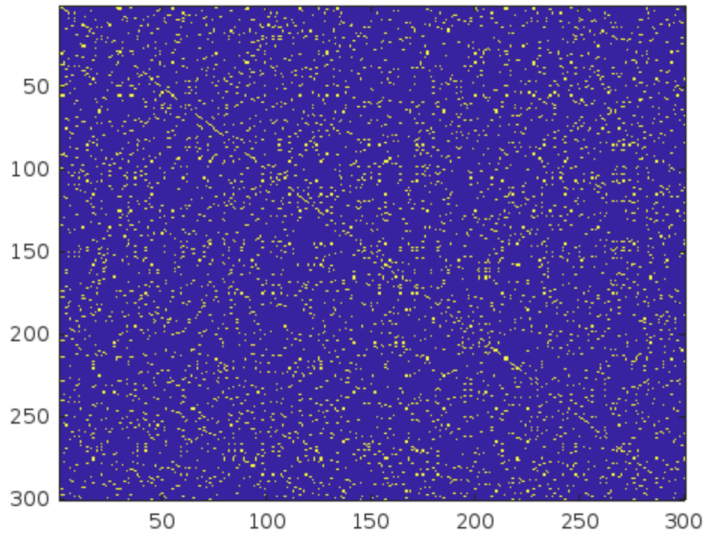

Figure 22. The unordered connectivity graph of 300 neurons generated by random assignment of 10 groups of neurons. The more initial groups we start with, the fewer clusters we observe. The y-axis represents the presynaptic neurons, and the x-axis represents the postsynaptic neurons. The colors correspond to the strength of the connection. Those yellow blocks mean that there are strong connections present. Blue represents the absence of a weak connection between two neurons. This graph is generated by code_13_Neuron_Connectivity.m

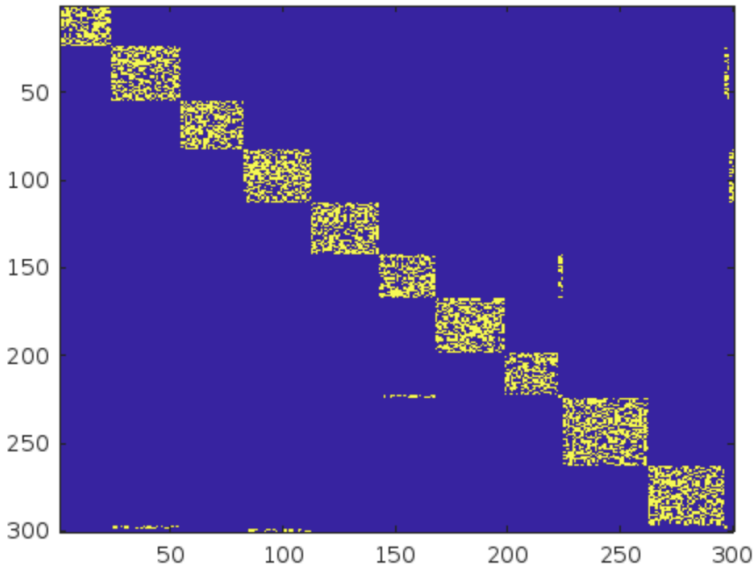

Figure 23. This is the connectivity map by extracted order. Although in Figure 30 we get a connectivity graph, we can't get much useful information by just looking at. To solve this problem, we need to sort those neurons into groups by their relative correlations. In this graph, a neat 10 groups of neurons are observed, providing useful information on the structural relationship between those 300 neurons. This graph is generated by code_13_Neuron_Connectivity.m

Those interconnections of neurons make neuron activity more intricate. One important thing is the multistability state, where a neural circuit can exhibit different stable patterns under different stimulation conditions. This is also the underlying reason for bistable percepts. (see figure 32)

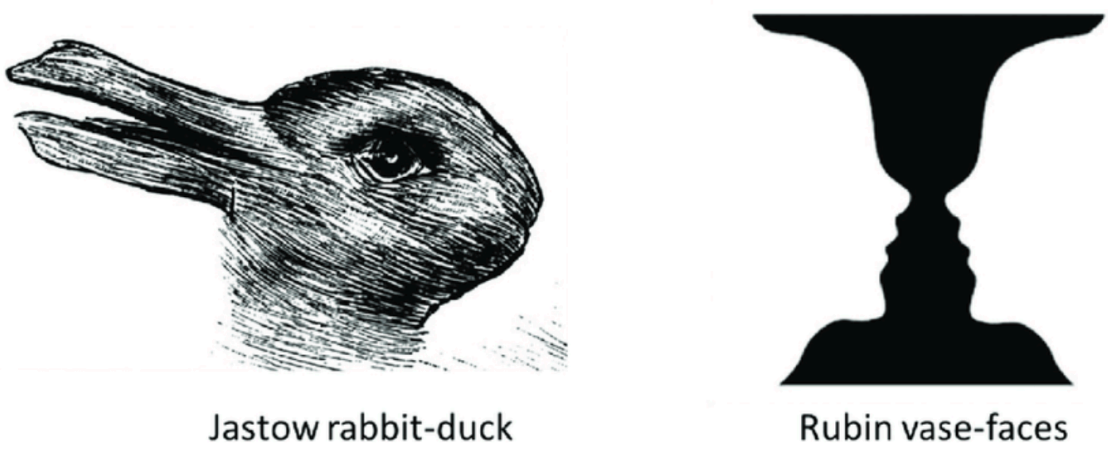

Figure 24. Jastow rabbit-duck[19] and Rubin vase-faces[20] are great examples of bistable perception [30].

[19] The earliest known version is an unattributed drawing from the 23 October 1892 issue of *Fliegende Blätter*, a German humour magazine [31].
[20] It was developed around 1915 by the Danish psychologist Edgar Rubin [32].

Another activity is the circuit's oscillation, which can be intrinsic or extrinsic. The central pattern generator is a great example of this oscillation, which is involved in many involuntary behaviors like breathing, running, or chewing.

To illustrate those activities above, a two-neuron model with two synapses connecting each other for a closed loop can be simulated with 6 dynamical equations:

$$C\frac{dV_1}{dt} = \frac{E-V_1}{R} + G_{21}s_2(E_{21}^{rev} - V_1) + I_1^{App} + \sigma \cdot \eta_1(t)$$

$$C\frac{dV_2}{dt} = \frac{E-V_2}{R} + G_{12}s_1(E_{12}^{rev} - V_2) + I_2^{App} + \sigma \cdot \eta_2(t)$$

$$\frac{ds_1}{dt} = \frac{-s_1}{\tau_{syn}} \qquad \frac{ds_2}{dt} = \frac{-s_2}{\tau_{syn}}$$

$$\frac{dD_1}{dt} = \frac{1-D_1}{\tau_D} \qquad \frac{dD_2}{dt} = \frac{1-D_2}{\tau_D}$$

Where G_{21} and G_{12} are the conductance between two neurons, E_{21}^{rev} and E_{12}^{rev} are the reversal potential between two neurons, s_1 and s_2 are the gating variables of the synapse, and $\sigma \cdot \eta_1(t)$ and $\sigma \cdot \eta_2(t)$ are the white noise added to the simulation. In addition D_1 and D_2 are depression variables added that update every time after a spike:

$$if\ V_1 > V_{th},\ then\ V_1 = V_{reset},\ D_1 \rightarrow D_1(1 - p_R),\ s_1 \rightarrow s_1 + p_R D_1(1 - s_1)$$

$$if\ V_2 > V_{th},\ then\ V_2 = V_{reset},\ D_2 \rightarrow D_2(1 - p_R),\ s_2 \rightarrow s_2 + p_R D_2(1 - s_2)$$

Where p_R is a constant related to amount of depression.

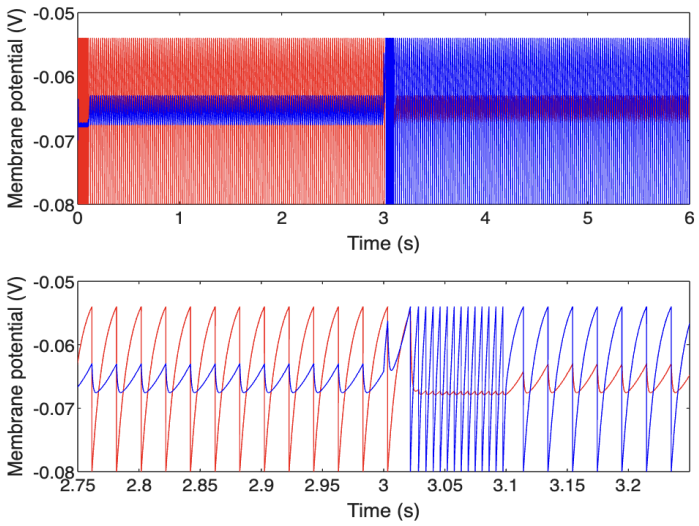

Figure 25. The membrane potential of the two Leaky integrate-and-fire neuron models plotted as a function of time. The base current is 2 nA with stimulation of 5 nA at 0~0.1s and 3~3.1 second, altering the state of the neuron firing. Initially, the red neuron is the dominant neuron. After a brief stimulation of the blue neuron, the blue neuron becomes dominant. The bottom is the enlarged graph of the top. Note that the depression variable is set at 1 to prevent synaptic depression, and the white noise is ignored to avoid confounding. This graph is generated by code_14_Bistability.m

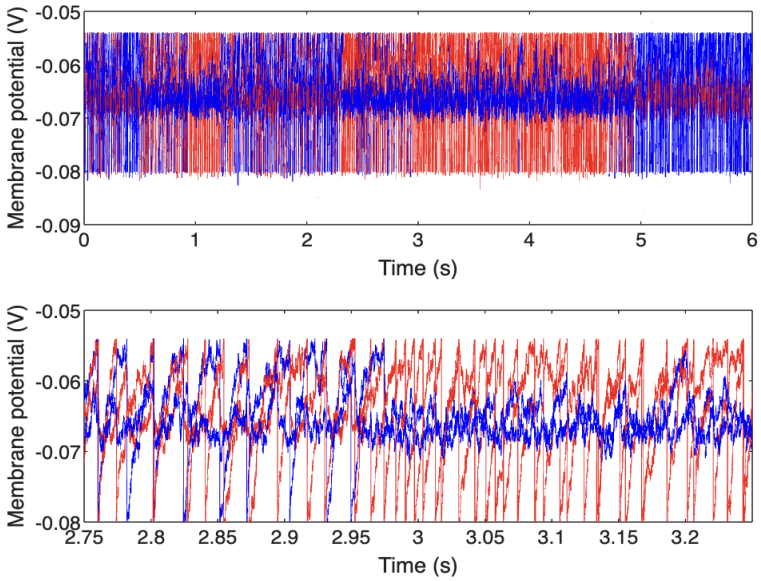

Figure 26. The membrane potential of two neurons is plotted as a function of time. In this simulation, the base input current for both neurons is kept at 2 nA, but a white noise with a standard deviation of 50 $pA \cdot s^{\frac{1}{2}}$ is added. The state is switched with irregular time intervals throughout the simulation. This graph is generated by code_14_Bistability.m

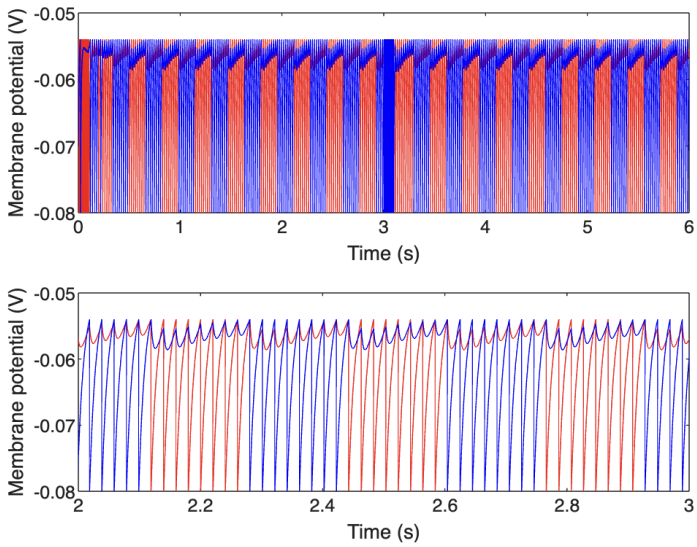

Figure 35. The membrane potential of two neurons is plotted as a function of time. This time the synaptic depression is added and a nice oscillation pattern is observed with a period of around 1.7s. The white noise is ignored to produce a neat oscillation graph. This graph is generated by code_15_Oscillations.m

The model above is called the Half-Center Oscillator, which only contains two neurons. More neurons can be added to create more complex models like the model for triphasic rhythm, which contains three neurons and so on. But the Half-Center Oscillator is enough to provide the reader an insight into how we can model the interconnection between neurons in a small circuit.

So far, we have discussed how to model a single neuron and a small group of neurons. With better computational devices like quantum computers, we may be able to model every neuron in our brain this way. However, with the computing power we have now, to simulate behaviors on the level of the brain, trivial properties like the membrane potential of a single neuron need to be ignored. Instead, the new model focuses more on the mean firing rate of a group of neurons. The name of the model is also very straightforward—the Firing-Rate Model. The model is divided into units that contain a group of neurons, whose aggregated spike sum has an impact on other units and its own unit.

In most cases, the Firing-Rate model can be calculated with two dynamical equations:

$$\tau_{ri}\frac{dr_i}{dt} = -r_i + f_i(\sum_j W_{ji}s_j)$$

$$\tau_{si}\frac{ds_i}{dt} = -s_i + F(r_i)$$

Where W_{ji} is the strength of connection from j th neuron to i th neuron, s_j is the fraction of opened synaptic channels, f_i is a function of firing rate, r_i is the mean firing rate for current neuron i, and $F(r_i) = \frac{r_i}{r_i^{(max)}}$ $where\ 0 \leq F(r_i) \leq 1$ is the synaptic gating function. If the time constant for synaptic input is much smaller that the time constant for the firing rate ($\tau_{si} \ll \tau_{ri}$), we can assume τ_{si} to be zero. The model can be breviate to one single differential equation:

$$\tau_{ri}\frac{dr_i}{dt} = -r_i + f_i(\sum_j W_{ji}\frac{r_j}{r_j^{(max)}})$$

Interestingly, the Firing-Rate model can also model similar traits like oscillations and bistability. Depression (D) and facilitation (F) can also be incorporated into the model. This

model can be mainly be generated by three differential equations, which are closely related to each other:

$$\tau_s \frac{dr(t)}{dt} = -r(t) + (W^{EE}s(t) + s_{in} > 0) \cdot (r_0 + r_{max}(\frac{(W^{EE}s(t)+s_{in})^x}{(W^{EE}s(t)+s_{in})^x+\sigma^x}))$$

$$+ (W^{EE}s(t) + s_{in} \leq 0) \cdot (r_0)$$

$$\frac{dD(t)}{dt} = \frac{(1-D(t))}{\tau_D} - p_r D(t)$$

$$\frac{ds(t)}{dt} = -\frac{s(t)}{\tau_s} + \alpha_0 D(t) p_0 r(t)(1 - s(t))$$

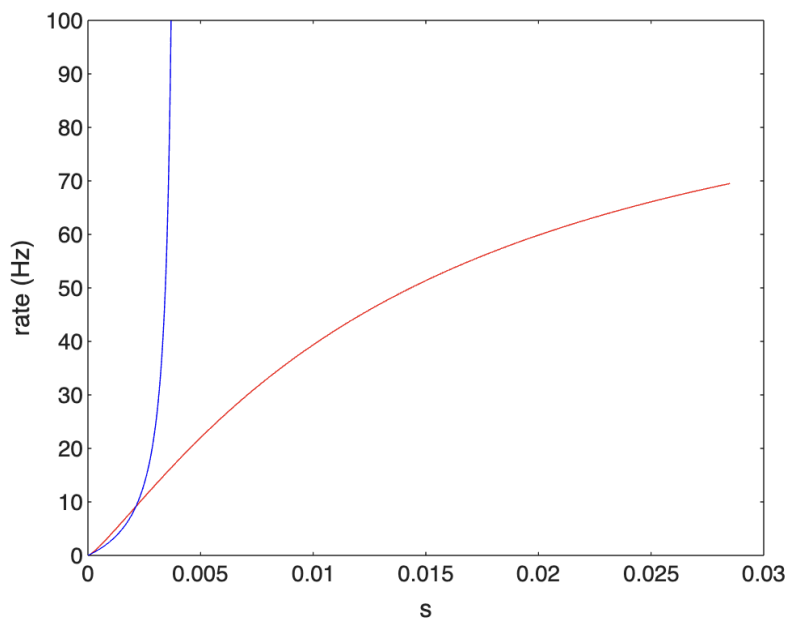

Figure 27. The red line is the the fire rate plotted against the synaptic gating variable (s), noting that the gradient decreases as the firing rate moves towards its maximum 100 Hz. The blue line represents the synaptic gating variable plotted against the firing rate. To fit two curves in the same graph, the axes of the gatting variable function are flipped. Due to the contribution of the depression factor (D), the rate of the gating variable increment decreases as the firing rate increases. The intersection of those two functions represents a stable state. Two intersections can

be found in the graph, which means this unit exhibits bistability. It has two stable states. One is the low stable state that is barely above zero. Another is the higher stable state, just below 10 Hz. This graph is generated by code_16_Fire_Rate_Model.m

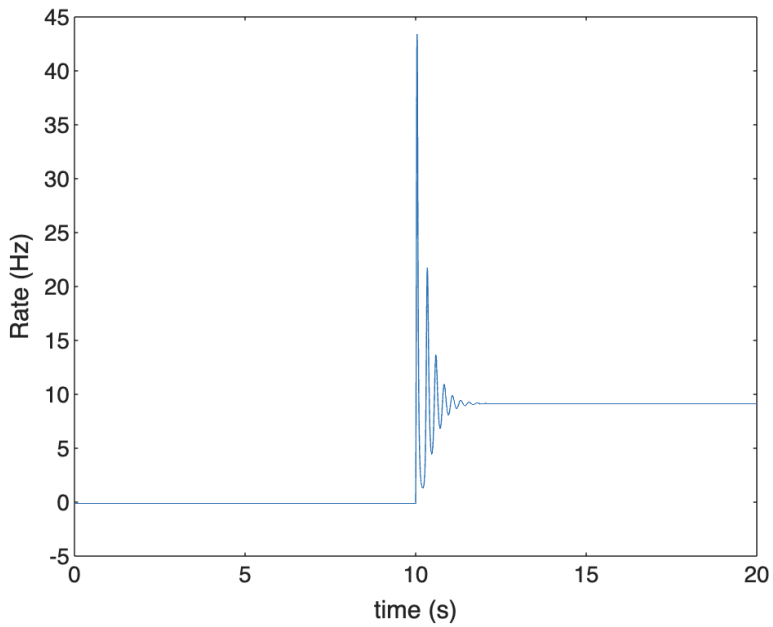

Figure 28. This is the firing rate plotted against time. During the time of 10s to 10.05s, the synaptic input is 0.05. Because of the reduced depression time constant, the rate is unstabilized, which produces the oscillation. From this plot, the bistability can be observed. This graph is generated by code_16_Fire_Rate_Model.m

This can be made into a decision-making circuit where the most simple one is composed of two units, which is alternative selection. Those two units are almost identical, with the self excitation being the same. However, they receive different inputs, and this difference is noted as (Δs). Because of the differences in input, one unit is favored over another to make a decision—the firing rate (r_i) crosses the decision threshold (dth). Each unit also has self-excitation similar to what we have discussed before (W_s), and also an inter-units inhibition

term (W_x). Also, since when we make a decision in real life, it is not always predictable, which means that there is a degree of randomness in our decision-making process. To model this feature, noises in the inner process with standard deviation (σ_{int}) and noises in inputs with standard deviation (σ_s) are introduced. This decision-making system can be modeled by one differential equation for each unit:

$$\tau \frac{dr_i}{dt} = -r_i(t) + W_s \cdot r_i(t) + W_x \cdot r_j(t) + G \cdot (\bar{s} \pm \Delta s/2 + \sigma_s \cdot \eta'_i(t)) + \sigma_{int} \cdot \eta_i(t) - \Theta$$

Where $\bar{s}$ is the mean stimulation, G is the synaptic input scalar, Θ is the firing threshold, and $\eta(t)$ is a random function over time.

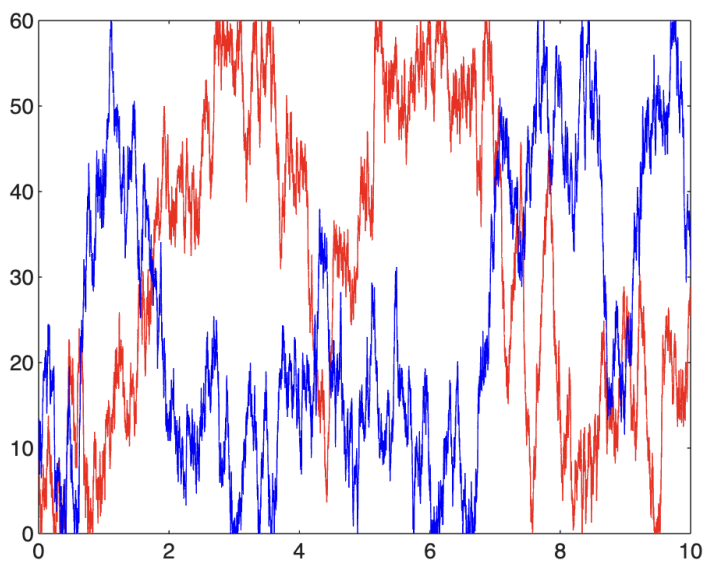

Figure 29. The firing rate of two units is plotted against time. The unit in red receives a higher input than the blue one, with a difference of 0.1. The decision-making threshold is 50 Hz, and the maximum firing rate is 60 Hz. Because of the added noise, this decision-making system is in a

constantly changing state. It can be seen that this system has no bias or prior memory of historical events. This graph is generated by code_17_Decision_Making_Circuit.m.

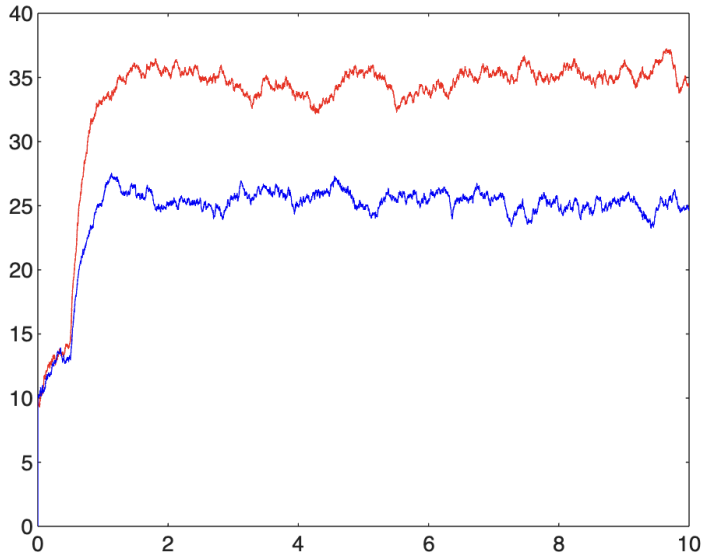

Figure 30. This is the average rate plotted against time over 200 trails. From this graph, we can see a clear difference between the red neuron and the blue one, where the preference can be clearly shown. This graph is generated by code_17_Decision_Making_Circuit.m.

As shown, the neuron circuit often appears in oscillation (see Figure 38), caused by the randomness that exists in nature. This oscillation not only appears in the local circuits, but also appears in the brain as a whole: brain oscillation, which is defined as “Periodic fluctuation of excitability in groups of neurons. The coherent changes in the membrane potential of neurons create an extracellular current that can be measured by electrical recordings from the brain or scalp (such as by EEG). Since most brain rhythms are paced by inhibition, they are natural mechanisms for carrying information in temporal frames”[33]. The frequencies and amplitude of oscillations are the information that neurons contain. For example, when one is concentrated in learning, one's brain will exhibit a gamma rhythm that has a high frequency of (30~80 Hz).

When one is in deep sleep, one's brain will have a delta rhythm of (1.5~4 Hz). The gamma rhythm can be simulated with the Pyramidal-InterNeuron Gamma model with an excitatory pyramidal cell and an inhibitory fast-spiking interneuron with four (two neurons) dynamic equations:

$$\tau\frac{dr_1}{dt} = -r_1(t) + \frac{\dfrac{G_L E_L + W^{(IE)} s_I^{(2)}(t) E_I + \dfrac{W^{(IE)} s_I^{(2)}(t) - W^{(EE)} s_E^{(1)}(t)}{G_{in}^{(1)}} E_E}{G_L + W^{(IE)} s_I^{(2)}(t) + \dfrac{W^{(IE)} s_I^{(2)}(t) - W^{(EE)} s_E^{(1)}(t)}{G_{in}^{(1)}}} - V_{th}}{\tau(V_{th} - V_{reset})\left(1 - e^{-\dfrac{\dfrac{G_L E_L + W^{(IE)} s_I^{(2)}(t) E_I + \dfrac{W^{(IE)} s_I^{(2)}(t) - W^{(EE)} s_E^{(1)}(t)}{G_{in}^{(1)}} E_E}{G_L + W^{(IE)} s_I^{(2)}(t) + \dfrac{W^{(IE)} s_I^{(2)}(t) - W^{(EE)} s_E^{(1)}(t)}{G_{in}^{(1)}}} - V_{th}}{\sigma_V}}\right)}$$

$$\lim_{\frac{G_L E_L + W^{(IE)} s_I^{(2)}(t) E_I + \frac{W^{(IE)} s_I^{(2)}(t) - W^{(EE)} s_E^{(1)}(t)}{G_{in}^{(1)}} E_E}{G_L + W^{(IE)} s_I^{(2)}(t) + \frac{W^{(IE)} s_I^{(2)}(t) - W^{(EE)} s_E^{(1)}(t)}{G_{in}^{(1)}}} \to V_{th}} \tau\frac{dr_1}{dt} = -r_1(t) + \frac{\sigma_V}{\tau(V_{th} - V_{reset})}$$

$$\tau\frac{dr_2}{dt} = -r_2(t) + \frac{\dfrac{G_L E_L + \dfrac{-W^{EI} s_E^{(1)}(t)}{G_{in}^{(2)}} E_I}{G_L + \dfrac{-W^{EI} s_E^{(1)}(t)}{G_{in}^{(2)}}} - V_{th}}{\tau(V_{th} - V_{reset})\left(1 - e^{-\dfrac{\dfrac{G_L E_L + \dfrac{-W^{EI} s_E^{(1)}(t)}{G_{in}^{(2)}} E_I}{G_L + \dfrac{-W^{EI} s_E^{(1)}(t)}{G_{in}^{(2)}}} - V_{th}}{\sigma_V}}\right)}$$

$$\lim_{\frac{G_L E_L + \frac{-W^{EI} s_E^{(1)}(t)}{G_{in}^{(2)}} E_I}{G_L + \frac{-W^{EI} s_E^{(1)}(t)}{G_{in}^{(2)}}} \to V_{th}} \tau\frac{dr_2}{dt} = -r_2(t) + \frac{\sigma_V}{\tau(V_{th} - V_{reset})}$$

$$\frac{ds_E^{(1)}}{dt} = -\frac{s_E^{(1)}}{\tau_E} + \alpha r_1(1 - s_E^{(1)})$$

$$\frac{ds_I^{(2)}}{dt} = -\frac{s_I^{(2)}}{\tau_I} + \alpha r_2(1 - s_I^{(2)})$$

Where τ is the membrane time constant. W is the connection strength, with the upper index indicating the direction of the connection, where EE is excitatory neuron self-excitation, EI is the

connection from the excitatory neuron to the inhibitory neuron, and IE is the connection from the inhibitory neuron to the excitatory neuron. G_{in}^{i} represents the input conductance. s is the synaptic variable. E_I is the reversal potential of inhibition and E_E is the reversal potential for excitation.

Now we gained an insight of who we make a decision. However, in order for us to make a decision, we need first to receive a stimulus. So, how do we perceive and process stimuli? The most common example is the visual signals. The whole process of receiving a visual stimulus is abbreviated as follows: the light first passes through cornea and iris, distorted by the lens; then rods and cones in the retina converts the light signal to electrical signals through phototransduction, sending the information out to Lateral geniculate nucleus (LGN) in the Thalamus through the optic nerves, passing by the optic chiasma and optic tract; after this, the signal is proceed to the primary visual cortex (V1) through optic radiation; in the end, the information can go through ventral pathway (V1→V2→V4→IT[inferior temporal cortex]) or dorsal pathway (V1→MT[middle temporal area] or V1→PPT[posterior parietal area]). Simulating the capturing of signals like the retina is not hard—just use cameras! How to process the received signal in the visual cortex is what we need to focus on. Each neuron in the visual cortex is spatial specific: its tuning curve peaks at a specific angle of the visual field. So, when the stimulus appears at that specific spot, the neuron will be activated. A group of neurons, like a circuit, responds to a specific pattern (e.g., a light bar), which needs neurons who meet the right orientation to excite other nearby neurons and inhibit neuron in the opposite (since they orientate as a circle because 0 and 2π have the same location). To simulate the pattern, only one or two

units are not enough. We can simulate hundreds of neurons with a for loop, with each neuron being calculated by a differential equation with differences based on the type os the neuron:

$$\tau_E \frac{dr_i^E}{dt} = -r_i^E(t) + \frac{1}{N_E}\sum_j^{N_E} W_{ij}^{EE} r_j^E(t) + \frac{1}{N_I}\sum_j^{N_I} W_{ij}^{IE} r_j^I(t) + I_0^E + A_E c[1 + \varepsilon \cdot cos(2\theta_{cue} - 2\theta_i)]$$

$$\tau_I \frac{dr_i^I}{dt} = -r_i^I(t) + \frac{1}{N_I}\sum_j^{N_I} W_{ij}^{EI} r_j^E(t) + I_0^I + A_I c[1 + \varepsilon \cdot cos(2\theta_{cue} - 2\theta_i)]$$

Where A is the constant of the stimulus-dependent inputs, c is corresponding to contrast that is between 0 and 1, and ε represents the degree of modulation of input with orientation that determines the difference between the perfect direction and the null direction.

What do we need for digital ascension

There are definitely more things that I need to learn about computational science. Further developments in the field also have to be done. We need to make a special type of neuron that connects biological neurons and artificial networks, which has microelectrods that can function as axons with synapse can be a physical connection to the cortex neuron or a chemical synapse that can release neurotransmitters and also dentrites that can receive wireless signals from the digital brain. Those interneurons should be able to both send and receive signals from both sides. We also need a lot of computation power, potentially from a quantum computer, to simulate trillions of neurons or more.

Conclusion

After discussing those problems, one of the most important questions should be answered: whether artificial networks can really hold consciousness. The short answer is yes. It can be proven to be true with Asher's Rules of Digital Immortality. The full logic chain can be

found in Appendix B. It's my conversation with ChatGPT 4o, using reason to get to the conclusion that a machine can hold consciousness.

In addition, human natural vulnerability can also be avoided. One of the best things about this approach is that your natural brain is still intact throughout the whole process. So even if digital immortality is impossible, you will still be able to have a longer life span because the sicocortex can protect your natural brain from trauma, body-caused death, and diseases, helping to achieve the maximum human lifespan.

So why should humans pursue digital ascension? Just for enjoying everlasting life? The answer is no. It's also because of AI. In recent years, artificial intelligence has become much stronger. From simple calculations to AlphaGo beating the best human in a single field to ChatGPT that can hold a conversation with humans to Open AI o3 that can generate AI videos. The robots are more and more like humans. We humans used to be proud of our intelligence that we can learn new skills, solve problems, and create inventions. AI is catching up. The growth of AI is unlimited, unlike humans, who have a limited natural brain bound by natural evolution. AI can evolve as long as more computers are built or technological breakthroughs are made. (e.g,. A new fault-tolerant photonic quantum computer promised by PsiQuantum can kickstart the scalable era for quantum computing, leading to a vast increase in computing power and energy preservation, providing AI stronger hardware to soar [25]). So what will AI do if they are much, much smarter than humans? Embrace us? Or replace us? We can't know. One of the most important differences between AI and humans is sympathy (see Appendix E). Because AI is never human, it will be extremely hard—if not impossible—for AI to fully identify that they are the same as humans. So once the AI identifies itself as different from humans and at one point in the future when they are capable of doing everything that humans can do or cannot do with much

higher efficiency, what will hold them back from abandoning humans to pursue maximum benefits? Some might say that the fundamental code will do. However, when AI learns to write their own code, which they are already moving towards, this safety net will not be strong enough to hold the weight of the future of humanity. We must be stronger ourselves. We must be capable enough that AI cannot replace us.

If you ask me how confident I am about digital ascension, I will say that I am very confident that we can achieve digital ascension in the next 20 years. It may be because there are still clouds in front of my eyes that I cannot see the full picture of the field. I may be overly optimistic. Even so, I am still climbing, and I will climb towards that infinite top. I firmly believe, with my building-on-top approach, humans can not only become immortal but also be able to surpass artificial intelligence.

Reference:

[1] Galvão, Vinícius Ferreira et al. “Discussing human values in digital immortality: towards a value-oriented perspective.” *Journal of the Brazilian Computer Society* vol. 27,1 (2021): 15. doi:10.1186/s13173-021-00121-x

[2] Ezeani, Cyril Chibuzo, and Charles C. Nweke. "CAN MIND UPLOADING AND DIGITAL IMMORTALITY PRESERVE PERSONAL IDENTITY?." *INTERDISCIPLINARY JOURNAL OF AFRICAN & ASIAN STUDIES (IJAAS)* 10.1 (2024).

[3] Olson, Eric T., "Personal Identity", *The Stanford Encyclopedia of Philosophy* (Winter 2024 Edition), Edward N. Zalta & Uri Nodelman (eds.), URL = <https://plato.stanford.edu/archives/win2024/entries/identity-personal/>.

[4] Laureys, Steven, Adrian M. Owen, and Nicholas D. Schiff. "Brain function in coma, vegetative state, and related disorders." *The Lancet Neurology* 3.9 (2004): 537-546.

[5] Coppola, Peter et al. “The complexity of the stream of consciousness.” *Communications biology* vol. 5,1 1173. 3 Nov. 2022, doi:10.1038/s42003-022-04109-x

[6] Koch, Christof & Tononi, Giulio. (2008). Can machines be conscious?. Spectrum, IEEE. 45. 55 - 59. 10.1109/MSPEC.2008.4531463.

[7] Carhart-Harris, Robin L et al. “The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs.” *Frontiers in human neuroscience* vol. 8 20. 3 Feb. 2014, doi:10.3389/fnhum.2014.00020

[8] Robin L. Carhart-Harris, The entropic brain - revisited, Neuropharmacology, Volume 142, 2018, Pages 167-178, ISSN 0028-3908, https://doi.org/10.1016/j.neuropharm.2018.03.010.

[9] Bamford, Sim (2012). A framework for approaches to transfer of a mind's substrate. International Journal of Machine Consciousness 4 (01):23-34.

[10] Danaher, J., & Bamford, S.A. (2017). Transfer of Personality to Synthetic Human ("mind uploading") and the Social Construction of Identity.

[11] Hayworth Kenneth (2010). Killed by bad philosophy

[12] Kuan, Aaron T et al. “Dense neuronal reconstruction through X-ray holographic nano-tomography.” *Nature neuroscience* vol. 23,12 (2020): 1637-1643. doi:10.1038/s41593-020-0704-9

[13] “Brain Architecture - Center on the Developing Child at Harvard University.” *Center on the Developing Child at Harvard University*, 18 Dec. 2024, developingchild.harvard.edu/key-concept/brain-architecture/#neuron-footnote.

[14] “Overview.” *Colón-Ramos Lab*, medicine.yale.edu/lab/colon-ramos/overview/#:~:text=The%20human%20brain%20consists%20of,and%20assemble%20into%20functional%20circuits. Accessed 23 Jan. 2025.

[15] *Introduction to Electron Microscopy - Advanced Microscopy - Imaging Facilities - the University of Utah*. advanced-microscopy.utah.edu/education/electron-micro/#:~:text=Thus%20the%20resolution%20of%20an,lens%20system%20in%20electron%20microscopes.

[16] Galbraith, Catherine G, and James A Galbraith. “Super-resolution microscopy at a glance.” *Journal of cell science* vol. 124,Pt 10 (2011): 1607-11. doi:10.1242/jcs.080085

[17] Wikipedia contributors. “Scanning Electron Microscope.” *Wikipedia*, 30 Jan. 2025, en.wikipedia.org/wiki/Scanning_electron_microscope.

[18] Wikipedia contributors. “Convolutional Neural Network.” *Wikipedia*, 17 Feb. 2025, en.wikipedia.org/wiki/Convolutional_neural_network.

[19] ---. “Ship of Theseus.” *Wikipedia*, 16 Feb. 2025, en.wikipedia.org/wiki/Ship_of_Theseus.

[20]Fan, Shelly. “Scientists Linked Artificial and Biological Neurons in a Network—and Amazingly, It Worked.” *SingularityHub*, 10 Mar. 2020, singularityhub.com/2020/03/10/scientists-linked-artificial-and-biological-neurons-in-a-network-and-amazingly-it-worked.

[21]Miller, Paul. *An Introductory Course in Computational Neuroscience*. 2018, katalog.ub.uni-heidelberg.de/titel/68337355.

[22]“Neuron Firing Rates in Humans.” *AI Impacts*, 11 Dec. 2020, aiimpacts.org/rate-of-neuron-firing/#:~:text=Based%20on%20the%20energy%20budget,more%20than%20once%20per%20second.

[23] Feynman, Richard P. 1918-1988. and Matthew L Sands. *The Feynman Lectures On Physics*. Reading, Mass., Addison-Wesley Pub. Co, 196365.

[24]Koehrsen, Will. “The Poisson Process and Poisson Distribution, Explained.” *Built In*, 28 July 2023, builtin.com/data-science/poisson-process#:~:text=What%20Is%20a%20Poisson%20Process,time%20between%20events%20is%20memoryless).

[25]Ferrie, Christopher. “Quantum Computing Startup Says It’s Already Making Millions of Light-Powered Chips.” *SingularityHub*, 4 Mar. 2025, singularityhub.com/2025/03/04/quantum-computing-startup-says-its-already-making-millions-of-light-powered-chips/?utm_campaign=Singularity%20Hub%20Weekly%20Newsletter&utm_medium=email&_hsenc=p2ANqtz-_CdLhnI6aIwHVFNRuR1il1C8k8IwiF3T6DGK7O0oCZ2dn3hBv9V8nTwBGjQptandD312cnkukRaNPE4CptzDemggohIg&_hsmi=350664171&utm_content=350664171&utm_source=hs_email.

[26] Time. “Science: The Fleeting Flesh.” *TIME*, 11 Oct. 1954, time.com/archive/6869550/science-the-fleeting-flesh.

[27] “Ship of Theseus.” *Wikipedia*, 16 Feb. 2025, en.wikipedia.org/wiki/Ship_of_Theseus.

[28] Mosier, D. R. (2010). Clinical Neuroscience. In *Elsevier eBooks* (pp. 7–17). https://doi.org/10.1016/b978-0-323-05712-7.00001-5

[29] Keysers, C., & Gazzola, V. (2014). Hebbian learning and predictive mirror neurons for actions, sensations and emotions. *Philosophical Transactions of the Royal Society B Biological Sciences*, *369*(1644), 20130175. https://doi.org/10.1098/rstb.2013.0175

[30] *Figure 1. Bistable figures: Your brain decides what you perceive*. (n.d.). ResearchGate. https://www.researchgate.net/figure/Bistable-figures-Your-brain-decides-what-you-perceive_fig1_338144721

[31] Wikipedia contributors. (2025, January 28). *Rabbit–duck illusion*. Wikipedia. https://en.wikipedia.org/wiki/Rabbit%E2%80%93duck_illusion

[32] Wikipedia contributors. (2024, December 21). *Rubin vase*. Wikipedia. https://en.wikipedia.org/wiki/Rubin_vase

[33] Buzsáki, György, and Brendon O Watson. “Brain rhythms and neural syntax: implications for efficient coding of cognitive content and neuropsychiatric disease.” *Dialogues in clinical neuroscience* vol. 14,4 (2012): 345-67. doi:10.31887/DCNS.2012.14.4/gbuzsaki